

BDA Course 2026

Lecture 3

Aki Vehtari, Osvaldo Martin

Outline Chapter 3

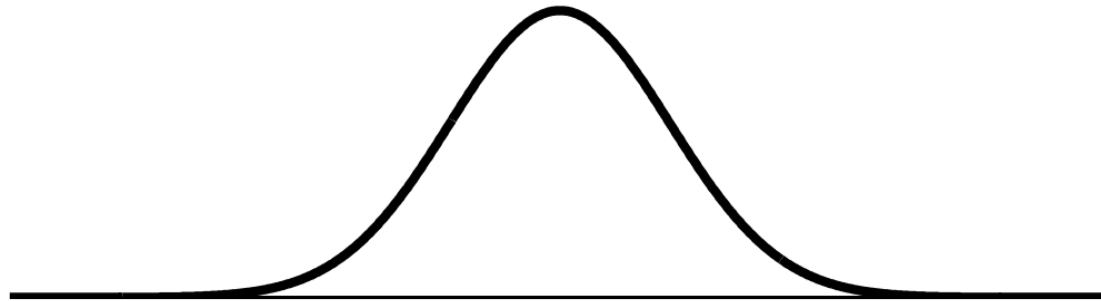
- 3.1 Marginalization
- 3.2 Normal distribution with a noninformative prior (important)
- 3.3 Normal distribution with a conjugate prior (important)
- 3.4 Multinomial model (can be skipped)
- 3.5 Multivariate normal with known variance (useful for chapter 4)
- 3.6 Multivariate normal with unknown variance (glance through)
- 3.7 Bioassay example (very important, related to one of the exercises)
- 3.8 Summary

This lecture

- Normal / Gaussian distribution with unknown mean and scale
- Marginalization by analytic integration
- Posterior predictive distribution by analytic integration
- Generalized Linear Models (GLMs)

Normal / Gaussian

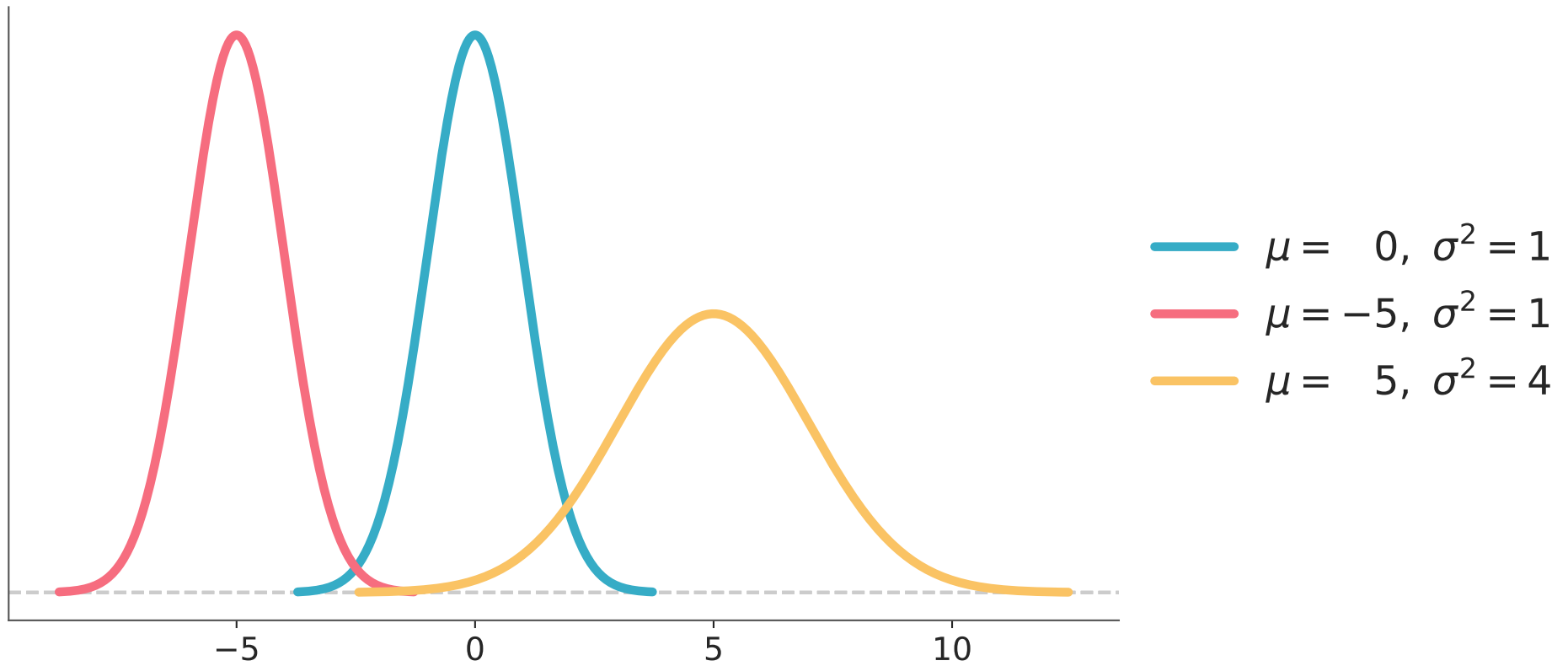
- Gaussian: favored by Germans (but it was studied earlier by De Moivre and Laplace, and Gauss avoided using it)
- Normal: some prominent statistician started to use this term systematically in the end of 19th century (but it's a bit of exaggeration)



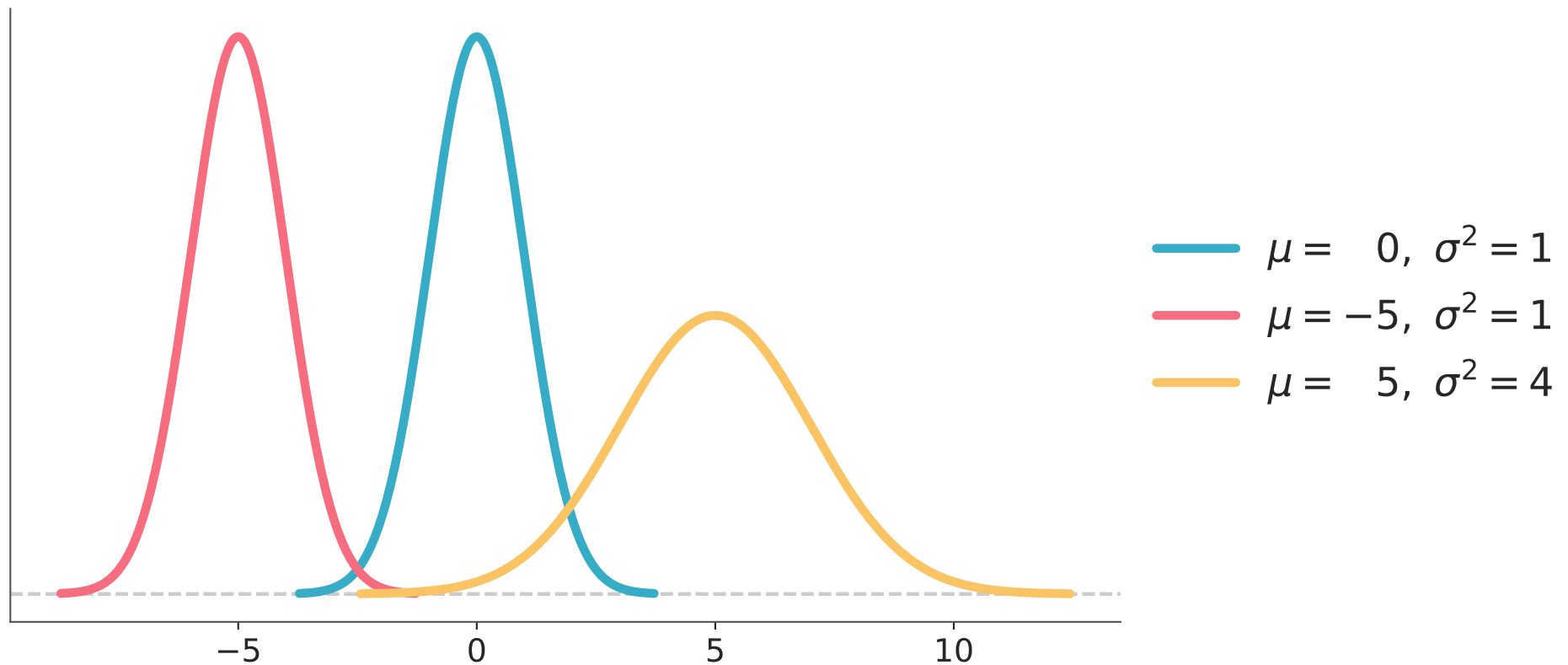
Normal / Gaussian

- On its own
 - Analysis of real valued observations
 - Sometimes convenient approximation for discrete observations
 - Often convenient assumption
- As a building block of more complex models
 - Normal linear regression
 - Gaussian processes (non parametric, infinite-dimensional generalization of the normal distribution)
 - Kalman filters (recursive estimation for linear Gaussian state-space models)
- As an approximation tool for Bayesian inference
 - Posterior distribution approximation with Laplace, variational inference, expectation propagation

Normal / Gaussian



Normal / Gaussian

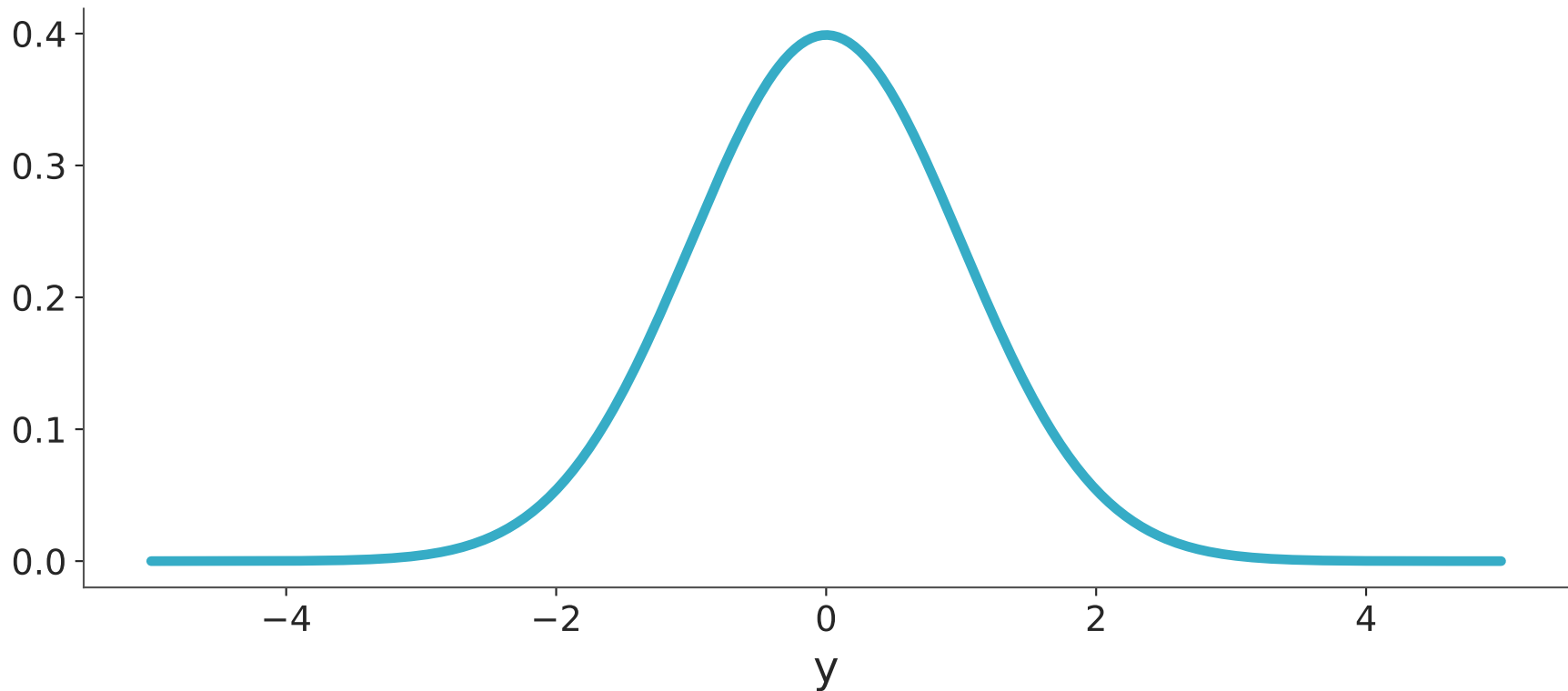


- $y \sim N(\mu, \sigma^2)$ with variance σ^2 (useful in derivations)
- $y \sim \text{normal}(\mu, \sigma)$ with deviation σ (useful for interpretation, used in many probabilistic programming languages, like Stan)

Normal / Gaussian

observation y , and parameters μ and σ

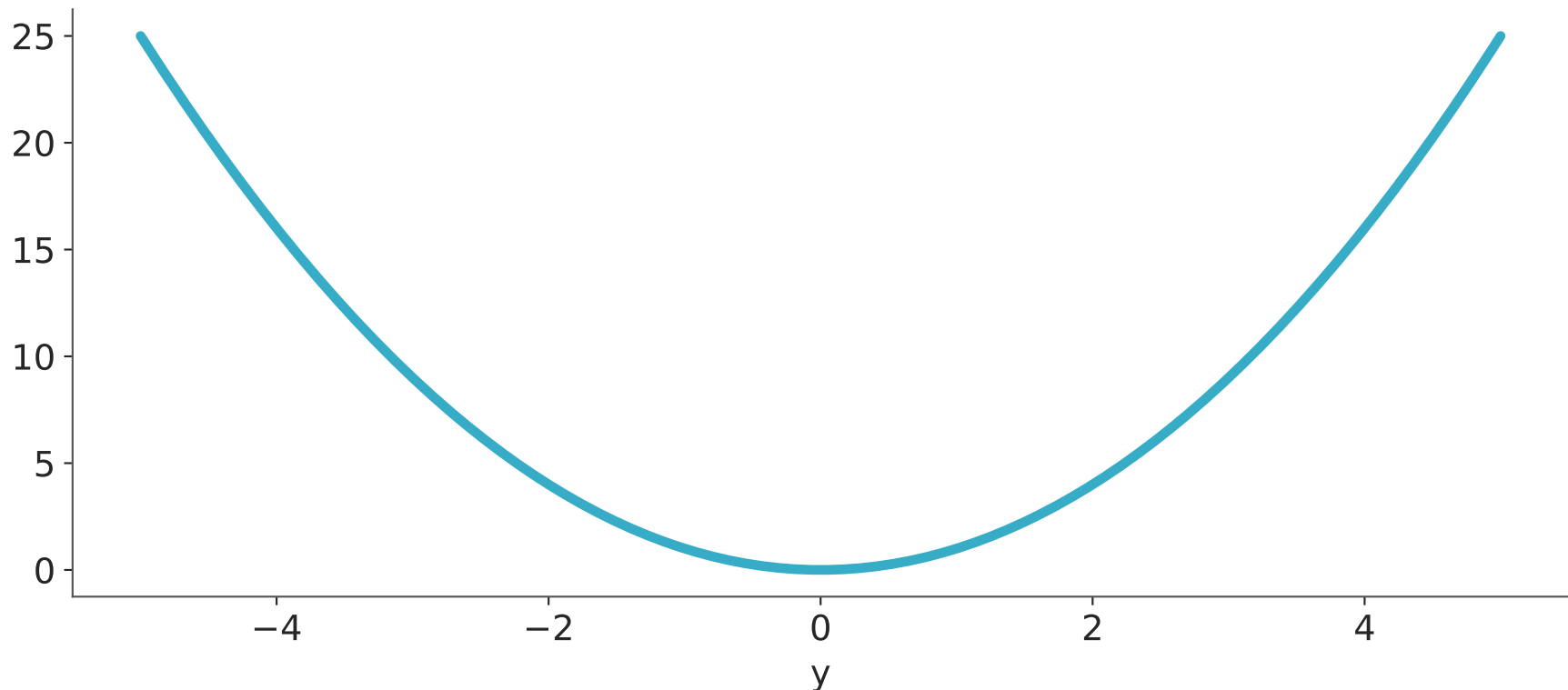
$$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$



Normal / Gaussian

- The square of the differences

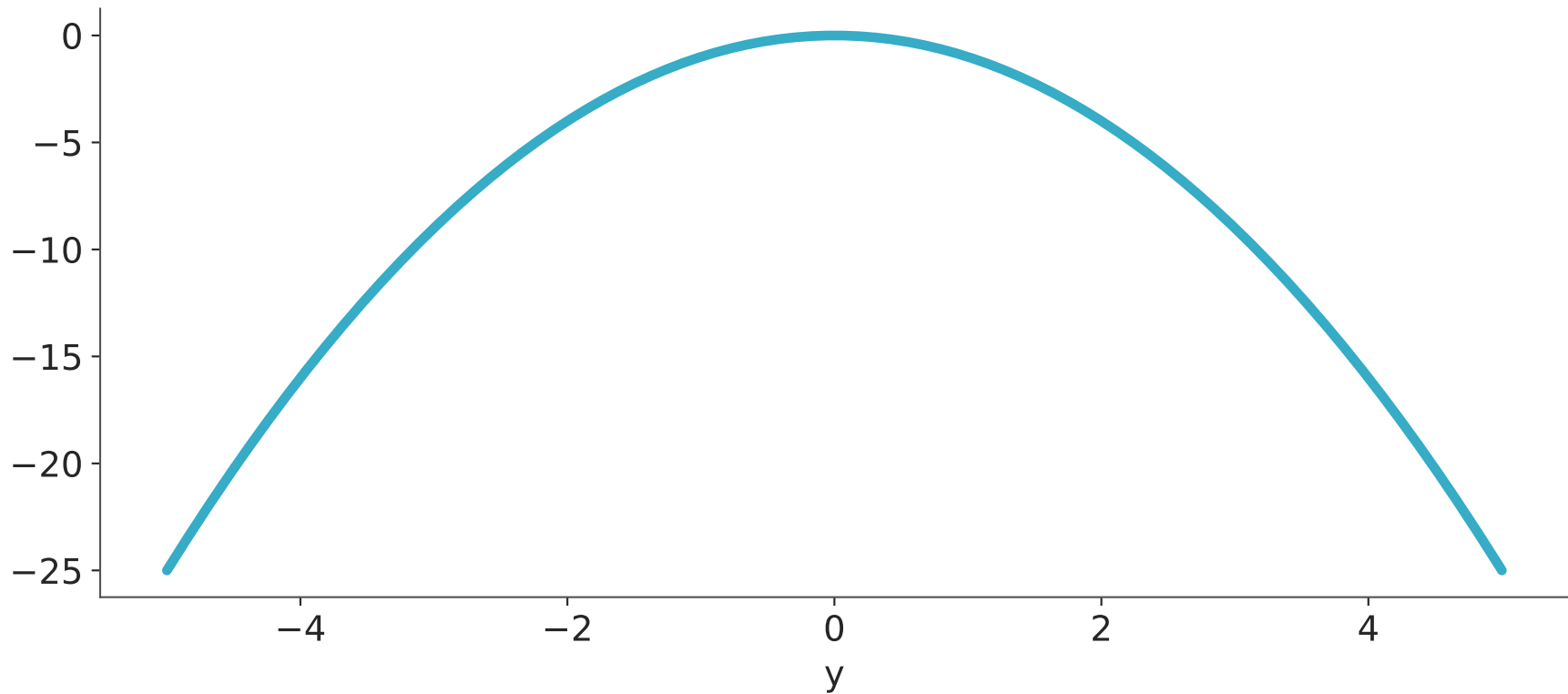
$$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$



Normal / Gaussian

- The minus square of the differences

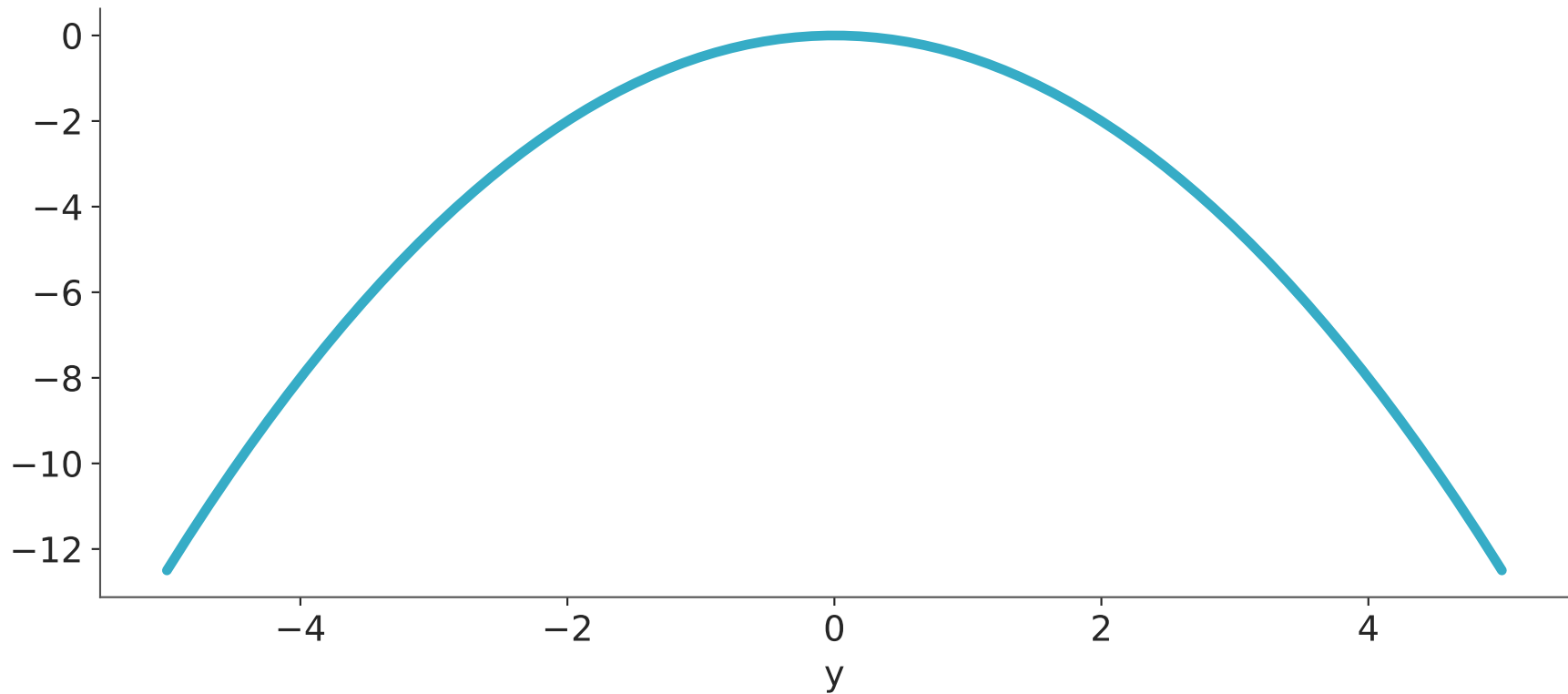
$$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$



Normal / Gaussian

- The scaled minus square of the differences

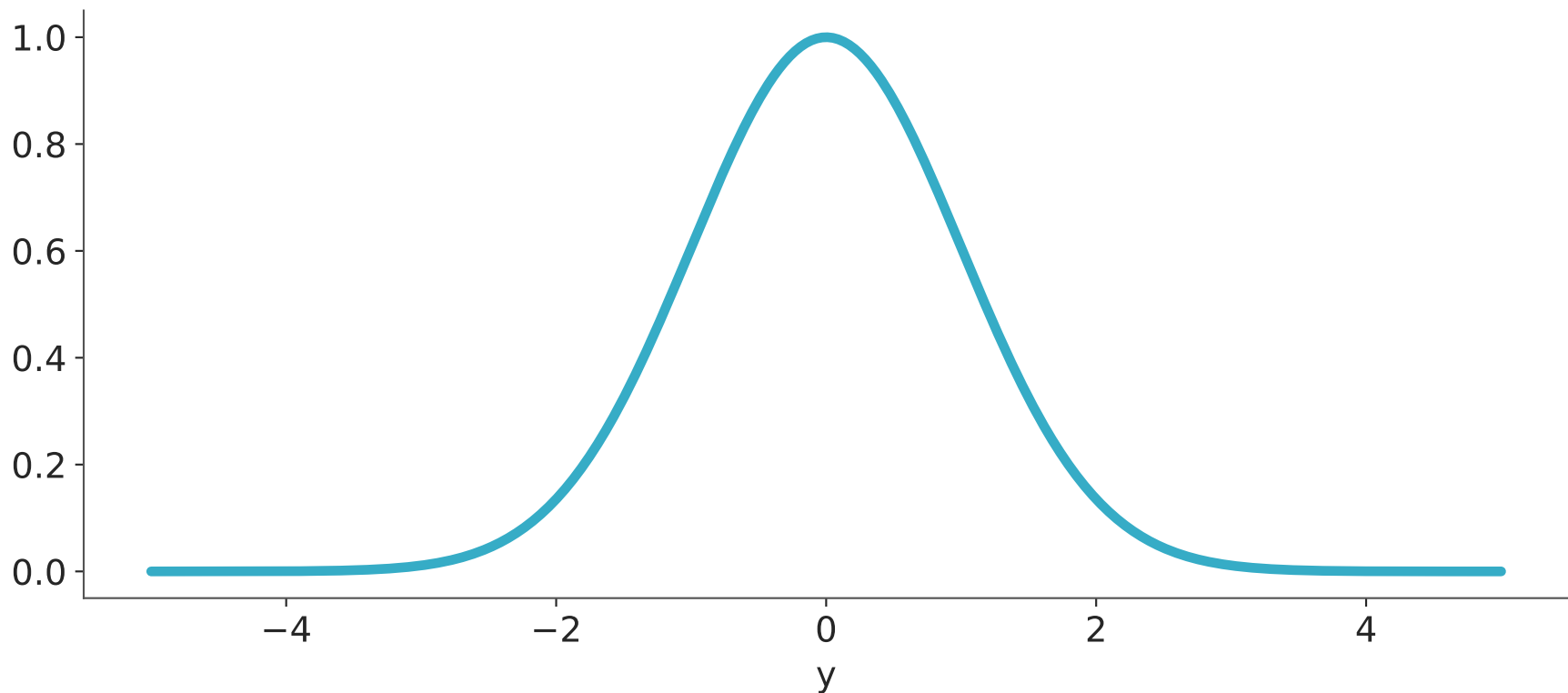
$$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$



Normal / Gaussian

- The exponentiated minus scaled square of the differences

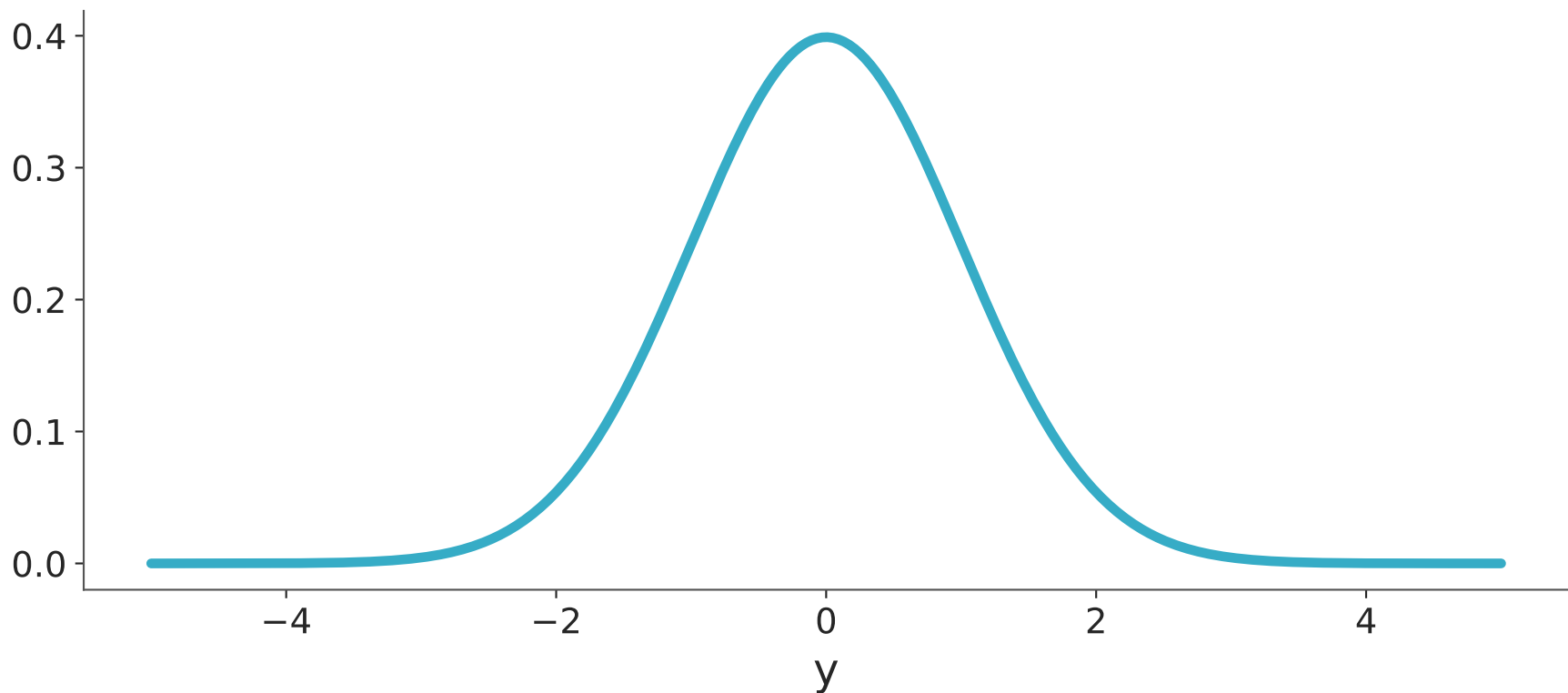
$$p(y \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$$



Normal / Gaussian

- The normalized exponentiated minus scaled square of the differences

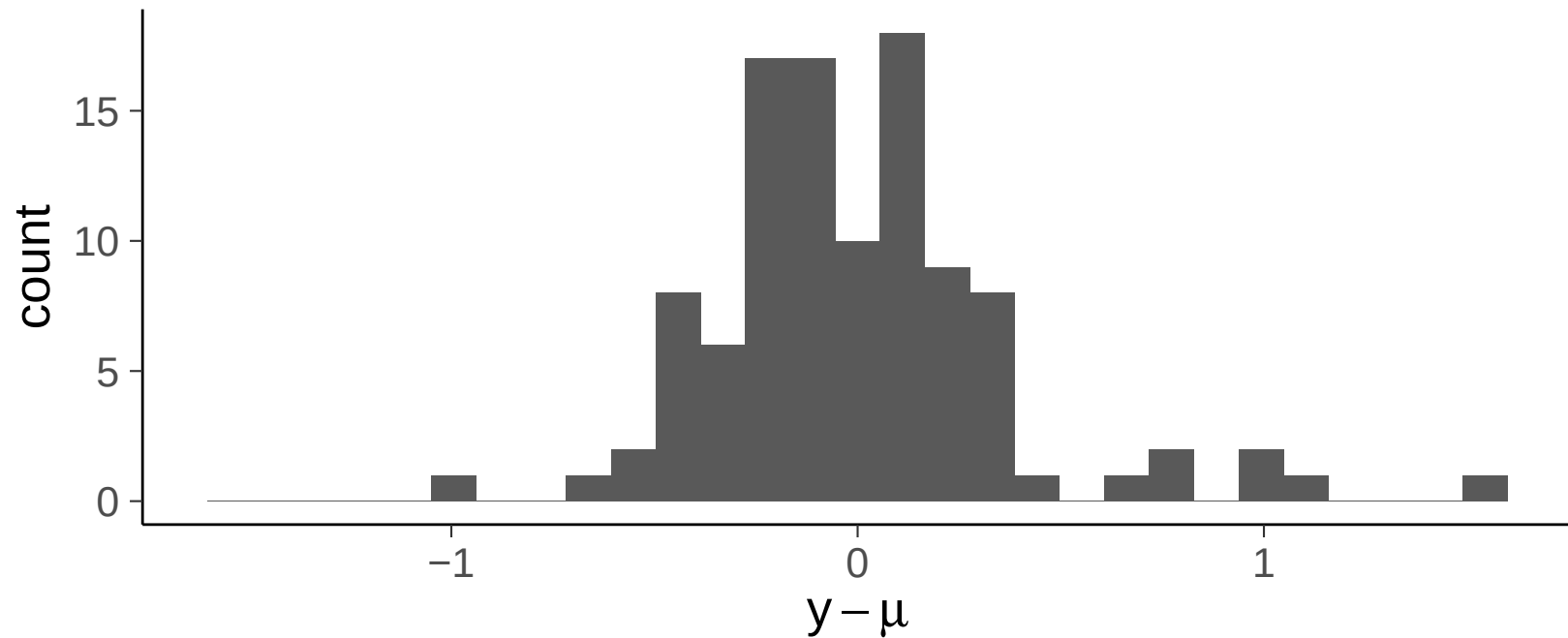
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Why π is here and how it is connected to CLT? → <https://www.youtube.com/watch?v=cy8r7WSuT1I>

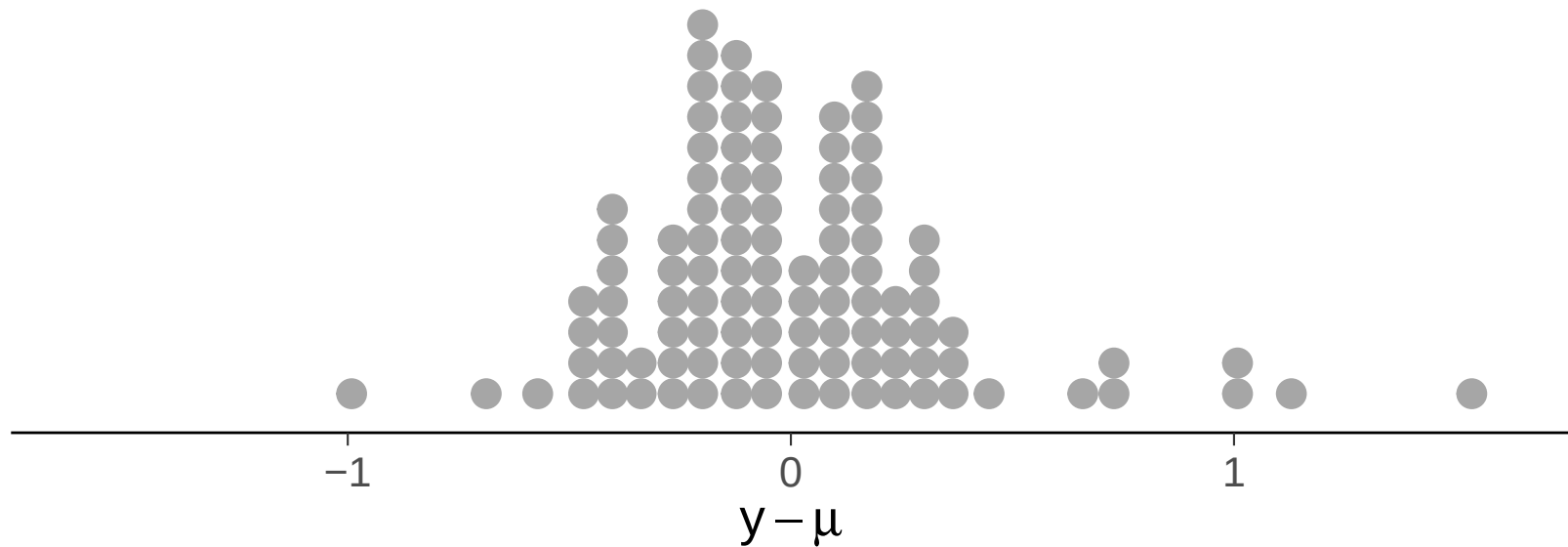
Variation in helicopter flight times

- Take the mean of the helicopter flight times
- To each observation, subtract the mean to get the residuals
- Histogram of residuals



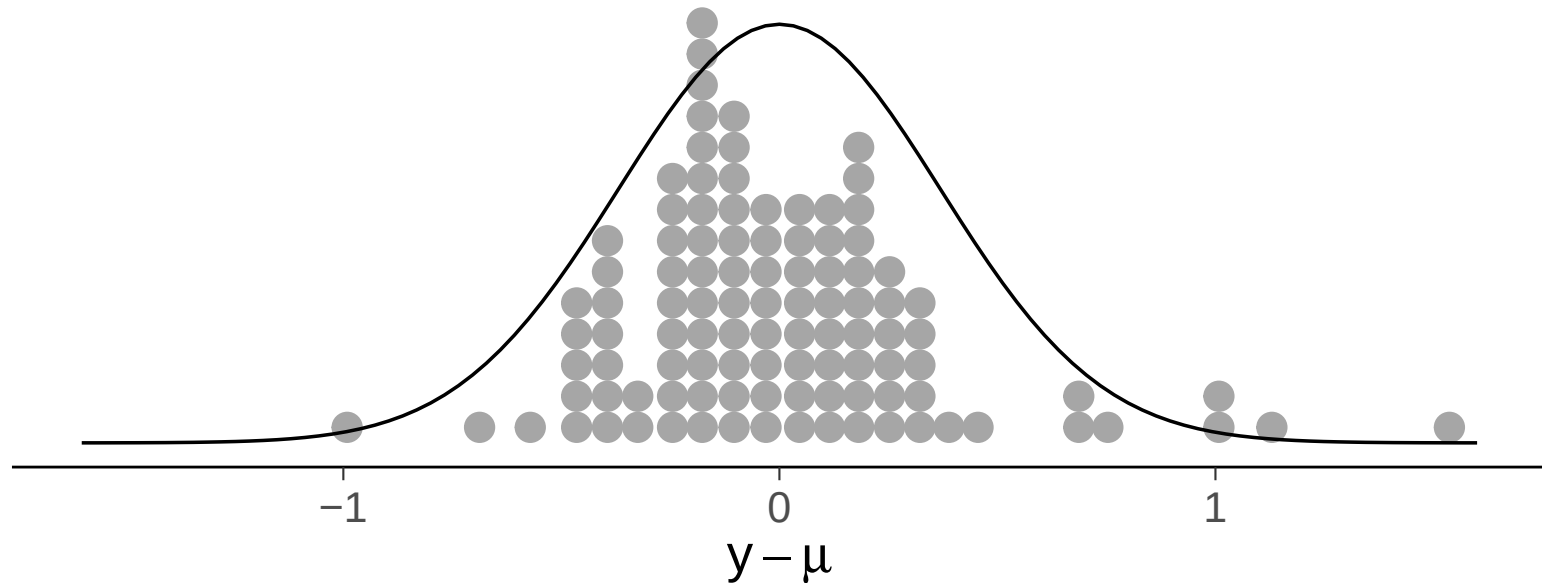
Variation in helicopter flight times

- Quantile-dot plot.
 - Each dot represents an equal fraction of the observations.
 - In this example we have 100 dots, so each dot is 1% of the observations.



Variation in helicopter flight times

- Quantile-dot plot + PDF of the normal distribution



- Why close to normally distributed?
 - Many small sources of variations summed together tend to be close to normally distributed
 - Central Limit Theorem (CLT)

Normal data model and likelihood

- **Data model** for y : $p(y \mid \mu, \sigma)$, a distribution over y for fixed μ, σ

Normal data model and likelihood

- **Data model** for y : $p(y \mid \mu, \sigma)$, a distribution over y for fixed μ, σ
- **Likelihood** the same expression $p(y \mid \mu, \sigma)$, now viewed as a function of μ, σ and fixed for y :

$$p(y \mid \mu, \sigma) = \prod_{i=1}^n p(y_i \mid \mu, \sigma),$$

with n helicopter flight times $y = (y_1, \dots, y_n)$ and assuming conditional independence.

Posterior for μ and σ given y

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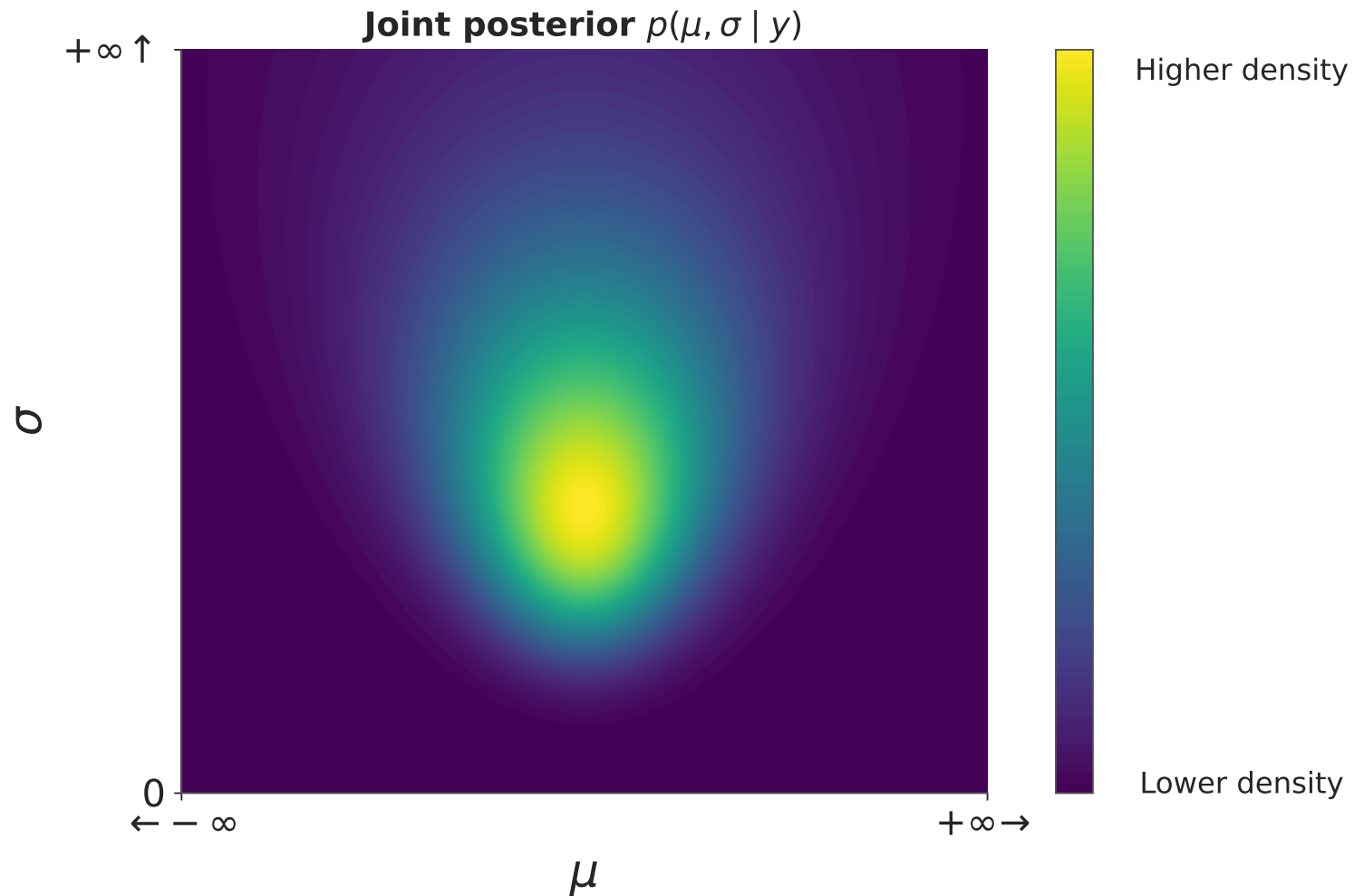
Posterior for μ and σ given y

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- **Prior** for μ and σ :
 - joint $p(\mu, \sigma)$
 - independent $p(\mu)p(\sigma)$
- Bayes' rule gives the joint **posterior**

$$p(\mu, \sigma \mid y) \propto \prod_{i=1}^n p(y_i \mid \mu, \sigma) \quad p(\mu, \sigma)$$

Posterior for μ and σ given y

- This posterior is 2-dimensional, covering both parameters jointly



- For some priors analytic solutions exist; for others we need numerical methods

Non-informative prior for μ, σ

- Assume independence:

$$p(\mu, \sigma) \propto p(\mu) \times p(\sigma)$$

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- We want $p(\sigma)$, not $p(\log \sigma)$. Change of variables (BDA3 p. 21):

$$p(\sigma) = p(\log \sigma) \left| \frac{d \log \sigma}{d\sigma} \right| = 1 \times \frac{1}{\sigma} = \sigma^{-1}$$

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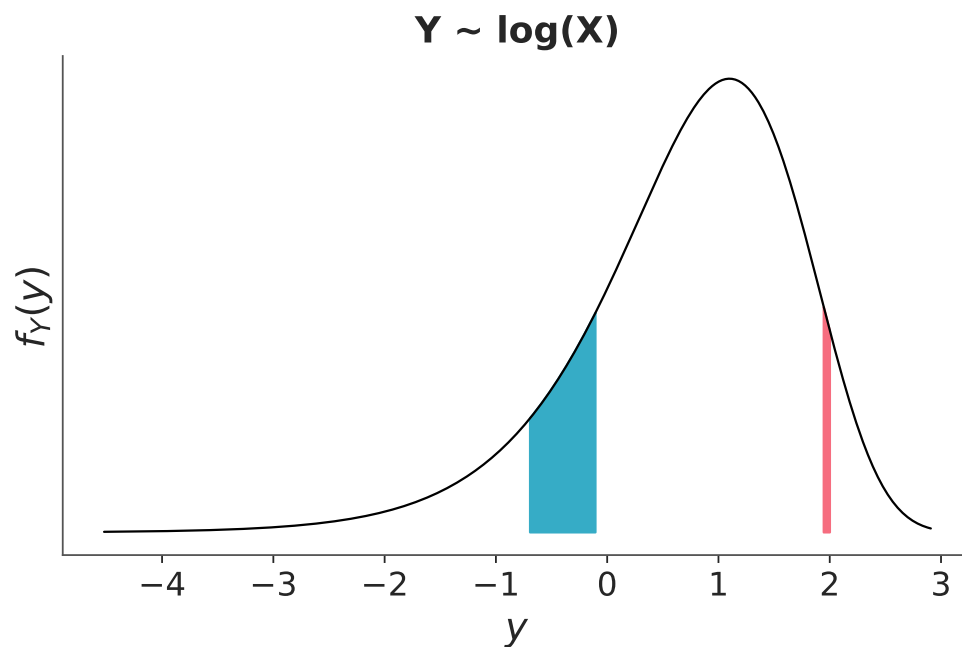
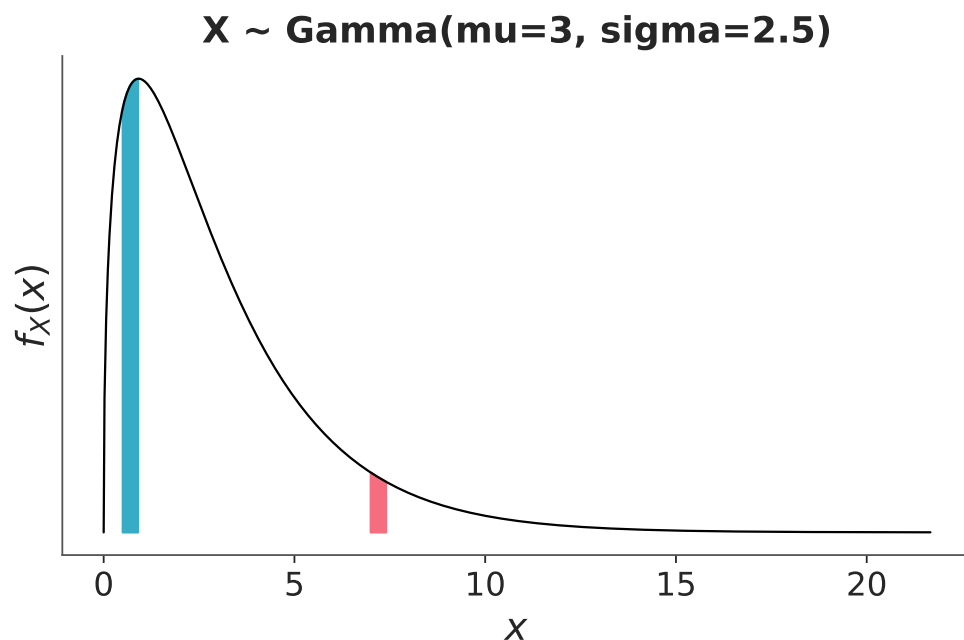
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- Putting it all together:

$$p(\mu, \sigma^2) \propto \sigma^{-2}$$

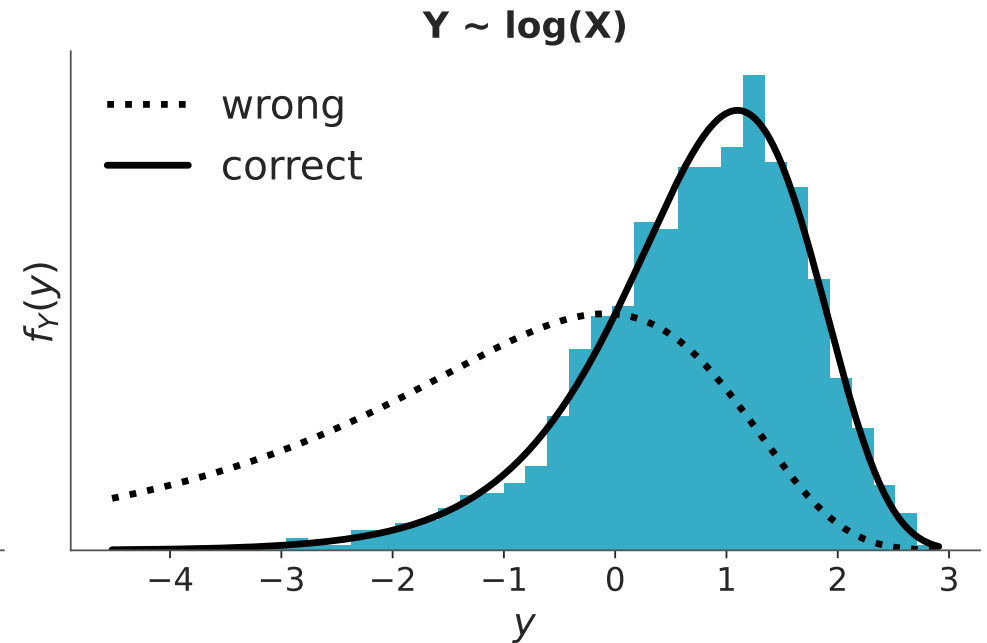
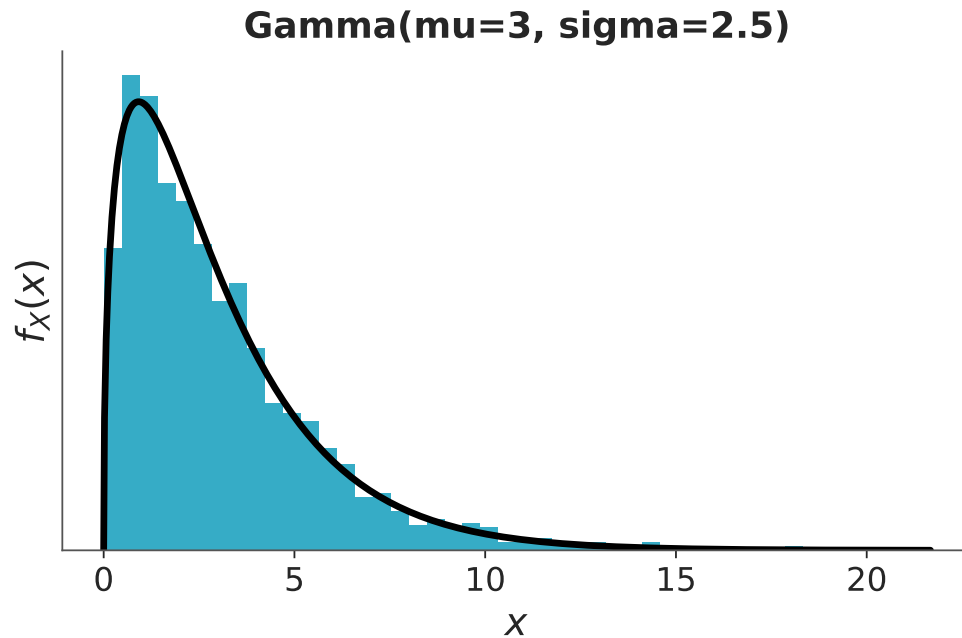
Change of variable intuition

- Suppose $X \sim \text{Gamma}(\alpha, \beta)$, but we want a new variable $Y = \log(X)$.
- The transformation squishes and stretches intervals, so the density must adjust to preserve probability.



Change of variable intuition

- A naive transformation $p_{X(\exp(Y))}$ gives the wrong density for $Y = \log(X)$.



Joint posterior

- Joint posterior

$$p(\mu, \sigma^2 \mid y) \propto \prod_{i=1}^n p(y_i \mid \mu, \sigma^2) \quad p(\mu, \sigma^2)$$

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$$p(\mu, \sigma^2 \mid y) \propto \prod_{i=1}^n p(y_i \mid \mu, \sigma^2) \quad p(\mu, \sigma^2)$$

- Substituting the non-informative prior and the normal density for each $p(y_i \mid \mu, \sigma^2)$:

$$p(\mu, \sigma^2 \mid y) \propto \sigma^{-2} \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mu)^2\right)$$

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- Using $\prod \frac{1}{\sigma} = \sigma^{-n}$ and $\prod \exp(a_i) = \exp(\sum a_i)$:

$$p(\mu, \sigma^2 \mid y) \propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2\right)$$

Joint posterior

$$\begin{aligned} p(\mu, \sigma^2 \mid y) &\propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2\right) \\ &\propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} \left[\sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2 \right]\right) \\ &\propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2]\right) \end{aligned}$$

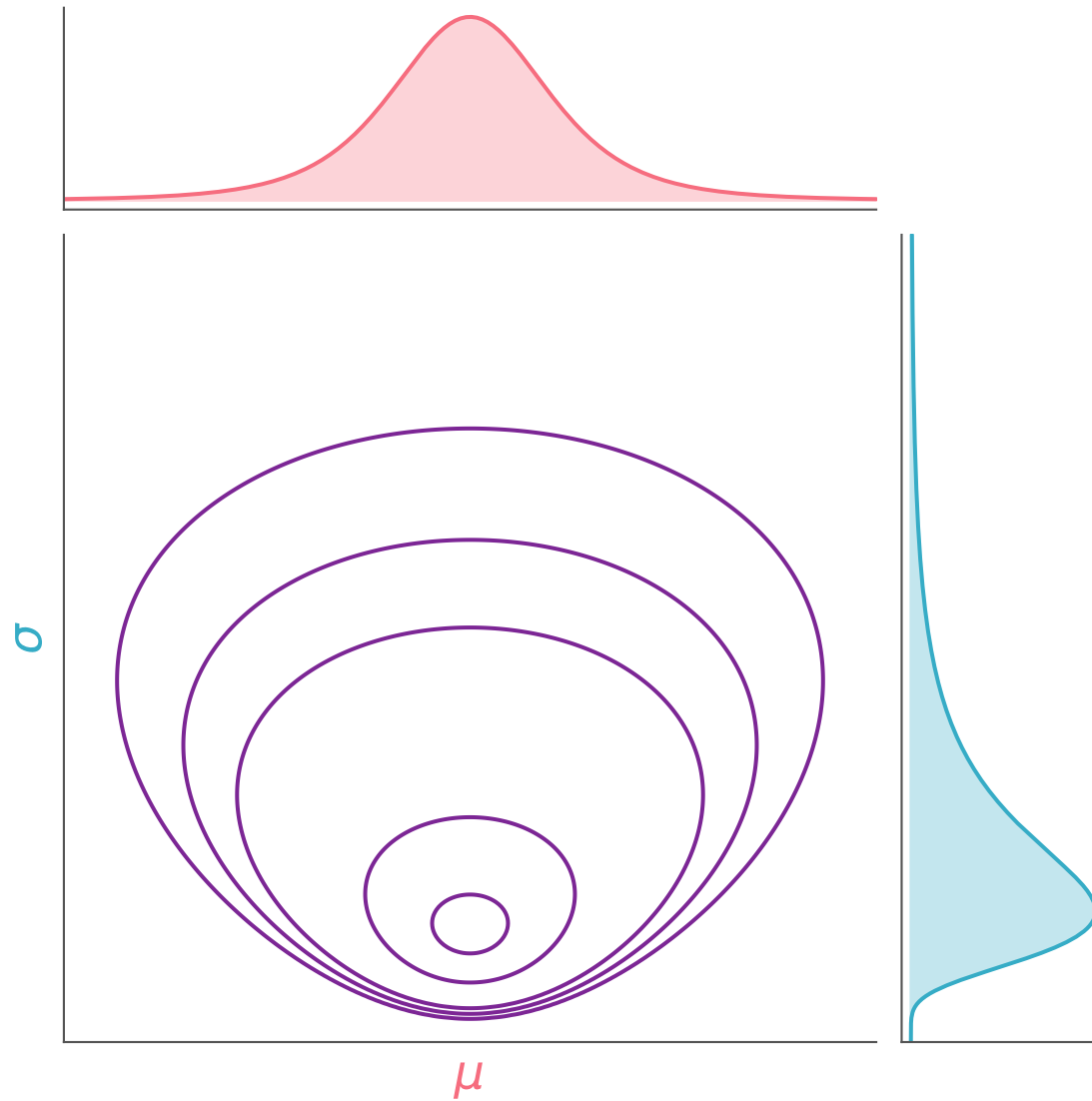
where $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$.

Step 1 → 2: split the sum into a part that does not depend on μ (the **within-sample** variation) plus a part that does (the **distance from the mean**). See appendix A.1.

Step 2 → 3: the first term is $(n-1)$ times the sample variance s^2

Takeaway: The only thing we need from the data is $\bar{y}$ and s^2 (sufficient statistics).

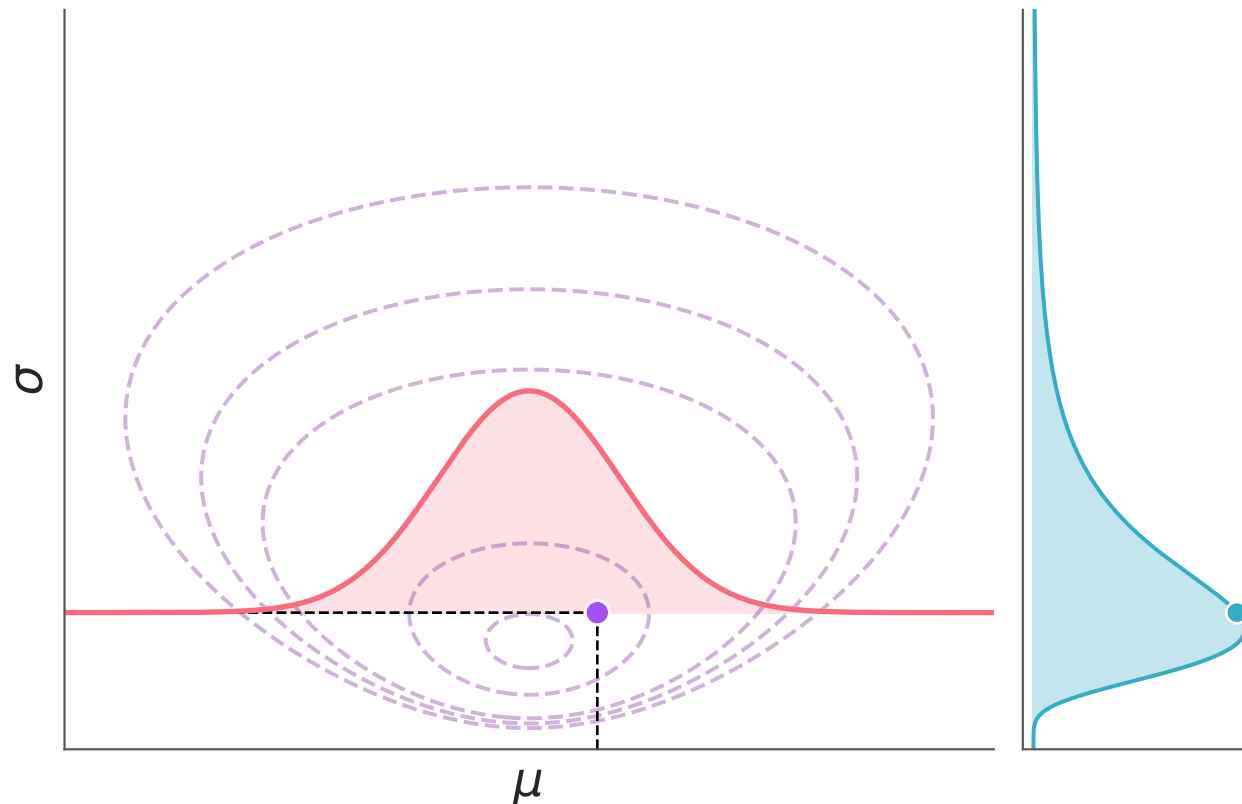
Joint posterior with marginals



Factorization

- Sampling from a **joint posterior** is often easier if done in steps:
 - 1. First draw from a **marginal**
 - 2. Then from a **conditional** given that draw.

$$p(\mu, \sigma^2 \mid y) = p(\mu \mid \sigma^2, y) p(\sigma^2 \mid y)$$



Marginal posterior $p(\sigma^2 \mid y)$, easier for σ^2 than σ

Integrate the joint posterior over the nuisance parameter μ :

$$\begin{aligned} p(\sigma^2 \mid y) &\propto \int p(\mu, \sigma^2 \mid y) \, d\mu \\ &\propto \int \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2]\right) \, d\mu \end{aligned}$$

Split the exponential ($e^{a+b} = e^a e^b$):

$$\propto \int \sigma^{-n-2} \exp\left(-(n-1)\frac{s^2}{2\sigma^2}\right) \exp\left(-\frac{n}{2\sigma^2}(\bar{y} - \mu)^2\right) \, d\mu$$

Only the second term involves μ .

$$\propto \sigma^{-n-2} \exp\left(-(n-1)\frac{s^2}{2\sigma^2}\right) \int \exp\left(-\frac{n}{2\sigma^2}(\bar{y} - \mu)^2\right) \, d\mu$$

Marginal posterior $p(\sigma^2 \mid y)$ (cont.)

The remaining integral is Gaussian in μ , with

$$\mu \mid \sigma, y \sim N\left(\bar{y}, \frac{\sigma^2}{n}\right).$$

Therefore,

$$\int \exp\left(-\frac{n}{2\sigma^2}(\bar{y} - \mu)^2\right) d\mu = \sqrt{\frac{2\pi\sigma^2}{n}} \propto \sigma.$$

Substituting back,

$$p(\sigma^2 \mid y) \propto \sigma^{-n-2} \cdot \sigma \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right)$$

and hence

$$p(\sigma^2 \mid y) \propto \sigma^{-(n+1)} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right).$$

Marginal posterior $p(\sigma^2 \mid y)$ (cont.)

Rewrite in terms of σ^2 :

$$p(\sigma^2 \mid y) \propto (\sigma^2)^{-\frac{n+1}{2}} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right).$$

This is the kernel of a scaled inverse- χ^2 distribution:

$$\sigma^2 \mid y \sim \text{Inv} - \chi^2(n-1, s^2)$$

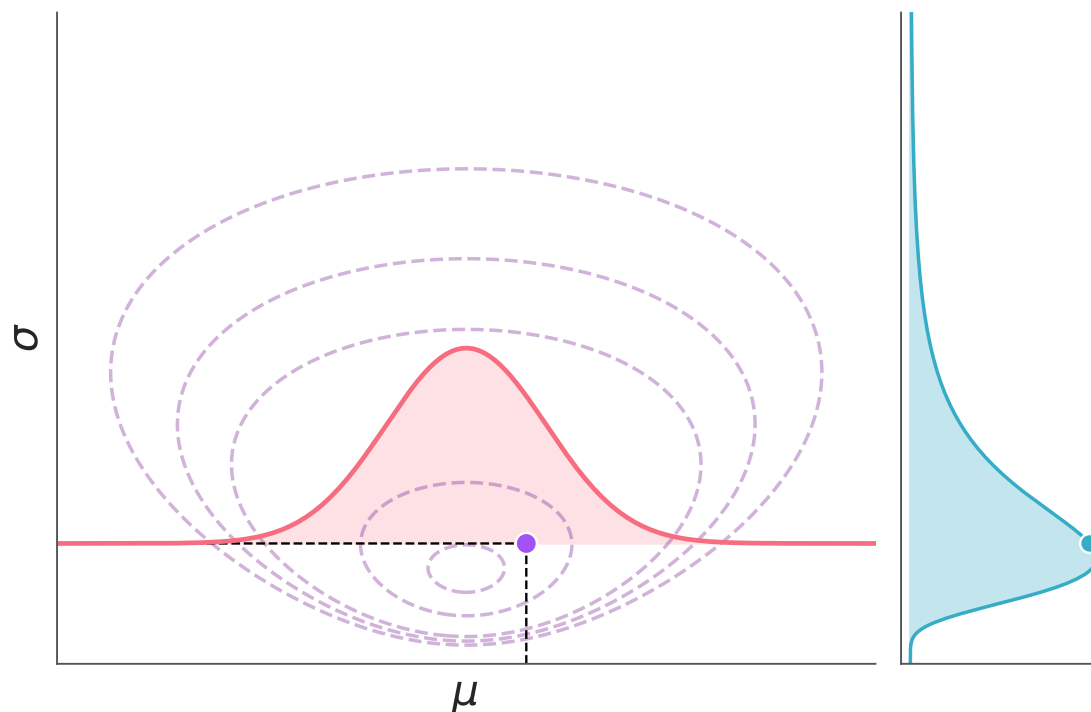
Factorization

- Sampling from a **joint posterior** is often easier if done in steps:
 - 1. Sample for the marginal:

$$p(\sigma^2 \mid y) = \text{Inv-}\chi^2(\sigma^2 \mid n - 1, s^2)$$

- 2. Given that draw, then from a **conditional** given that draw.

$$p(\mu \mid \sigma^2, y) = N\left(\bar{y}, \frac{\sigma^2}{n}\right)$$



Factorization

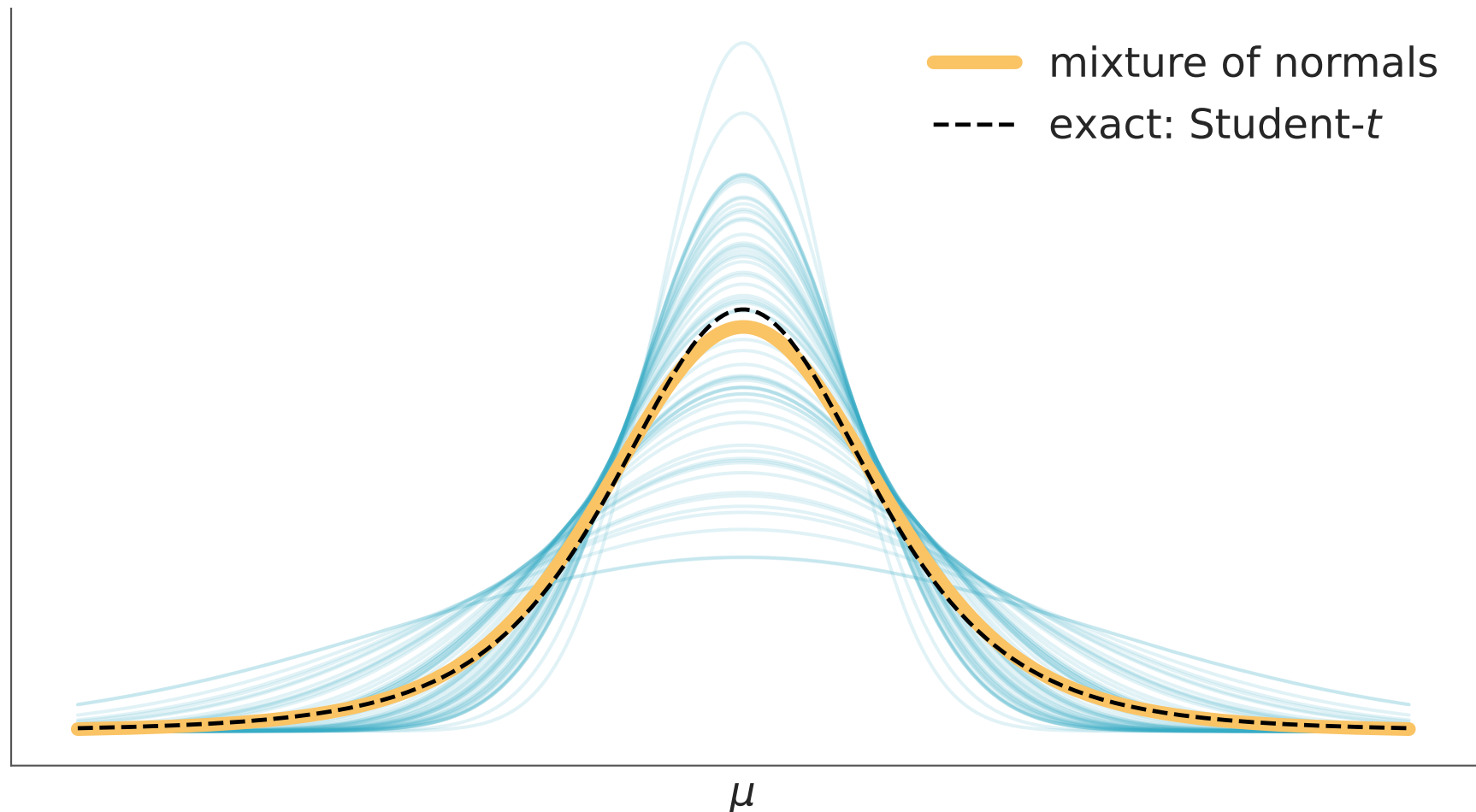
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Factorization

- Repeating for many draws of (σ^2) $\rightarrow$ a **family of normals** (one per draw)
- Averaging over all these conditional normals approximates the marginal posterior for μ (**a mixture of normals**):
- This mixture is in fact a **Student- t distribution** (See appendix A.2)



Predictive distribution for new $\tilde{y}$

- We often want to predict a **new** observation $\tilde{y}$

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- This requires averaging the **data model** over our **uncertainty in μ, σ** .

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \mu, \sigma) p(\mu, \sigma \mid y) d\mu d\sigma$$

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- Step 1: draw parameters from the posterior

$$\mu^s, \sigma^s \sim p(\mu, \sigma \mid y)$$

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- Step 1: draw parameters from the posterior

$$\mu^s, \sigma^s \sim p(\mu, \sigma \mid y)$$

- Step 2: draw a new observation from the data model

$$\tilde{y}^s \sim p(\tilde{y} \mid \mu^s, \sigma^s)$$

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- Step 2: draw a new observation from the data model

$$\tilde{y}^s \sim p(\tilde{y} \mid \mu^s, \sigma^s)$$

- Step 3: repeat steps 1 and 2 many times to approximate the predictive distribution

Posterior predictive distribution (Normal with known variance)

We integrate the data model over the conditional posterior for μ :

$$p(\tilde{y} \mid \sigma^2, y) = \int p(\tilde{y} \mid \mu, \sigma^2) p(\mu \mid \sigma^2, y) d\mu$$

$$p(\tilde{y} \mid \sigma^2, y) = \int N(\mu, \sigma^2) N\left(\bar{y}, \frac{\sigma^2}{n}\right) d\mu$$

The integral of a normal against a normal is again normal:

$$= N\left(\bar{y}, \sigma^2 + \frac{\sigma^2}{n}\right)$$

The *extra* $\frac{\sigma^2}{n}$ comes from not knowing μ exactly.

Posterior predictive distribution

In practice σ^2 is **not** known, so we also integrate over its posterior uncertainty:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \sigma^2, y) p(\sigma^2 \mid y) d\sigma^2$$

Integrating over the uncertainty in σ^2 gives a Student- t :

$$p(\tilde{y} \mid y) = t_{n-1} \left(\bar{y}, \left(1 + \frac{1}{n} \right) s^2 \right)$$

Compared with $p(\mu \mid y)$:

$$\frac{s^2}{n} \quad \text{vs.} \quad \left(1 + \frac{1}{n} \right) s^2$$

Predicting a new observation is less certain than estimating μ : it includes parameter uncertainty + observation noise

Normal - conjugate prior

- There are conjugate priors for the normal model
 - Check Section 3.3 in BDA3 for details.
 - Wikipedia has a list of [conjugate priors](#) for various models
- Useful sometimes when speed and analytical convenience (or insight) are important.
- With modern methods like MCMC, conjugate priors are less critical for computation.

Comparison of means of two normals

- The difference of two normally distributed variables is normally distributed
- The difference of two t distributed variables with different variances and degrees of freedom doesn't have a closed form
 - but easy to sample from the two distributions, and obtain draws of the differences

$$\text{if } \mu_1^s \sim p(\mu_1 \mid y_1)$$

$$\mu_2^s \sim p(\mu_2 \mid y_2)$$

$$\delta^s = \mu_1^s - \mu_2^s$$

$$\text{then } \delta^s \sim p(\delta \mid y_1, y_2)$$

- This is related to Monte Carlo method that will be discussed in the next lecture.

Normal linear regression

- We allow μ to depend on predictors x_i through a linear model:

$$y_i \sim N(\alpha + \beta x_i, \sigma^2), \quad i = 1, \dots, N$$

- For some case we can have analytical solutions for the posterior
- For the general case with arbitrary priors and unknown σ^2 , numerical methods are needed (discussed later in the course).
- more about regression in BDA3 Chapter 14 (not required for the course) and [Regression and Other Stories](#) book

Generalized linear model (GLM)

- We generalized regression models to allow for non-normal observation models.

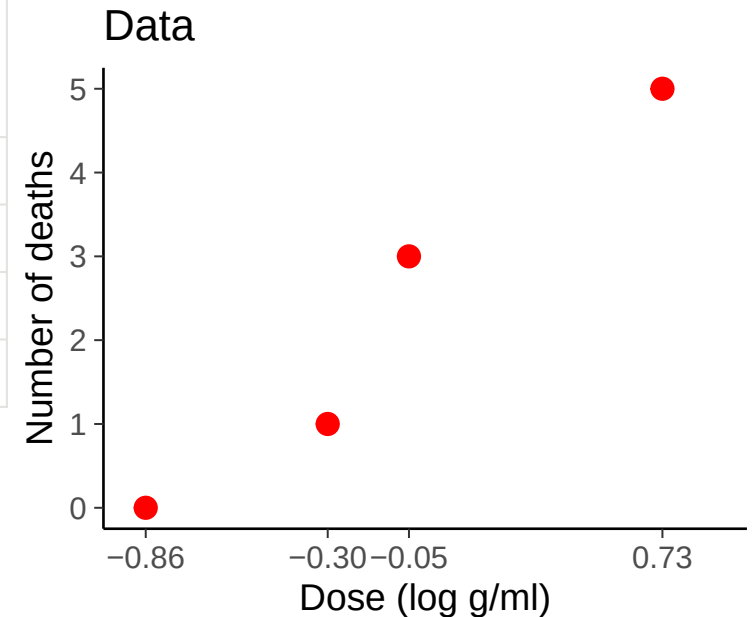
$$y_i \sim p(g^{-1}(\eta_i), \theta)$$

$$\eta_i = \alpha + \beta x_i$$

- We need a link function g to relate the linear predictor η_i to the mean (or other relevant parameter) of the observation model.
 - identity for normal
 - logit for Bernoulli
 - log for Poisson
 - ...
- θ represents other parameters of the observation model (e.g., variance for normal)
- More in BDA3 Chapter 16 and [Regression and Other Stories](#) book

Bioassay

Dose x_i (log g/ml)	Number of animals n_i	Number of deaths y_i
-0.86	5	0
-0.30	5	1
-0.05	5	3
0.73	5	5



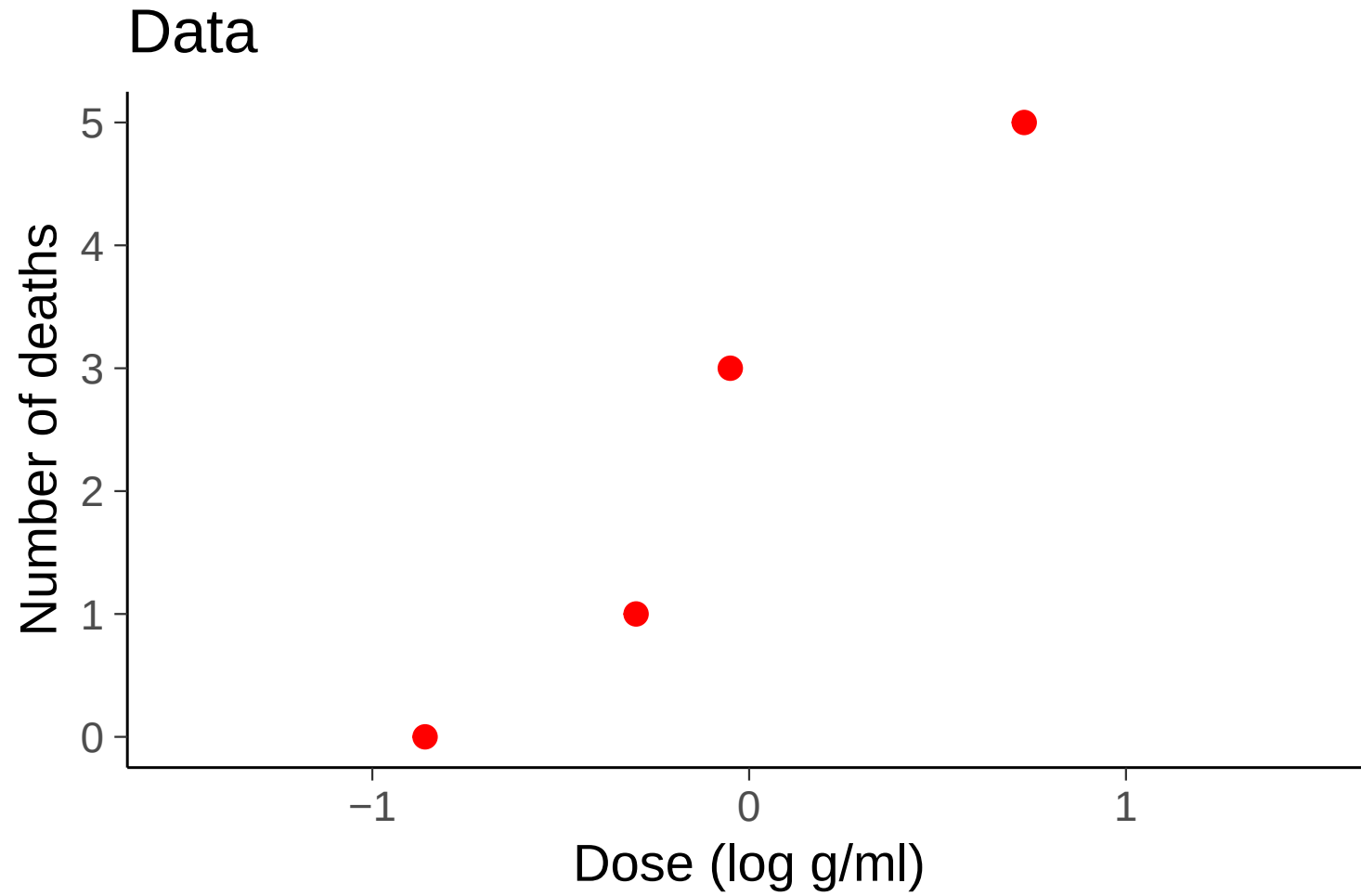
Find out lethal dose 50% (LD50)

- used to classify how hazardous chemical is
- 1984 EEC directive has 4 levels (see the chapter notes)

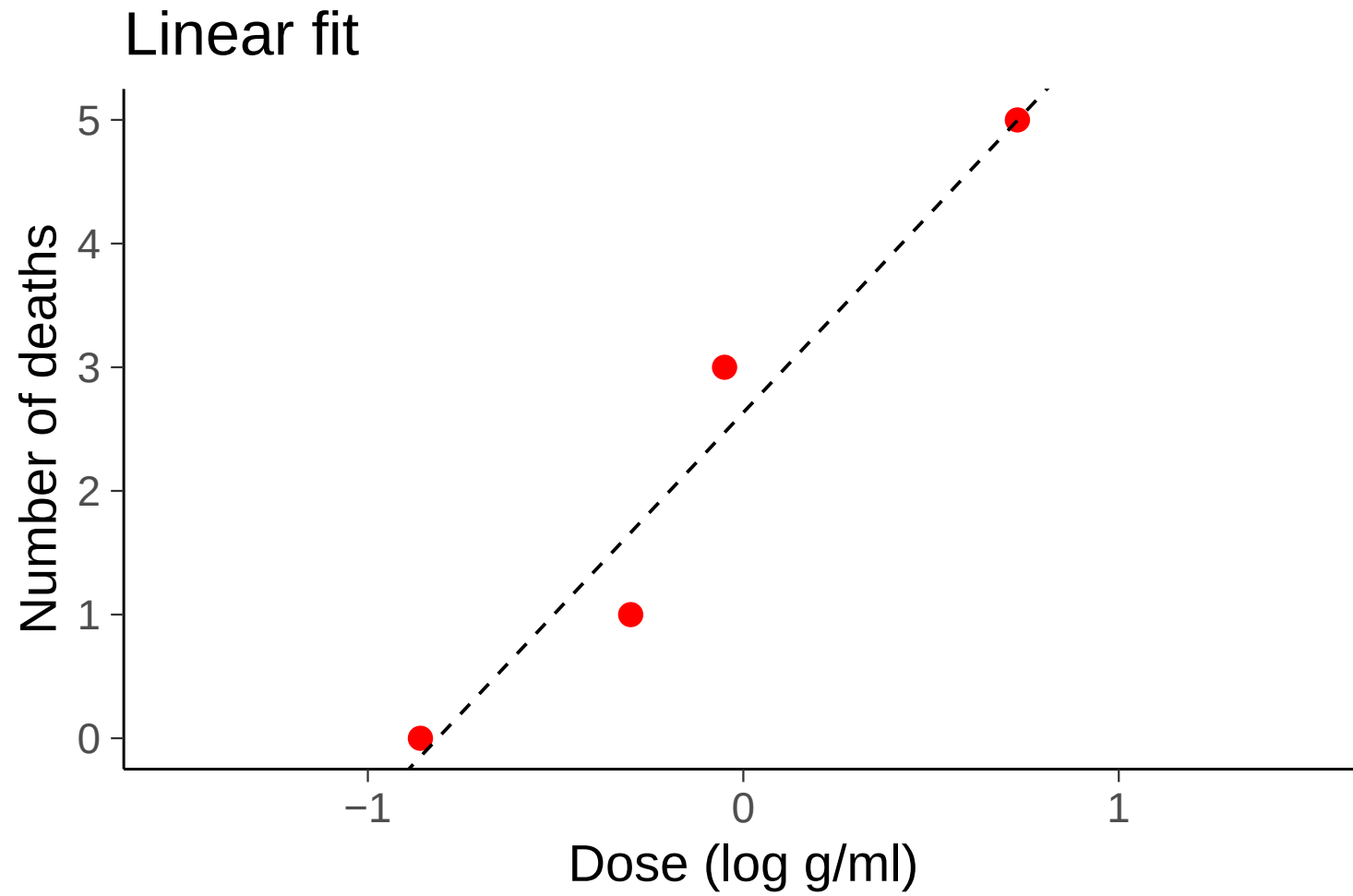
Bayesian methods help to

- reduce the number of animals needed
- easy to make sequential experiment and stop as soon as desired accuracy is obtained

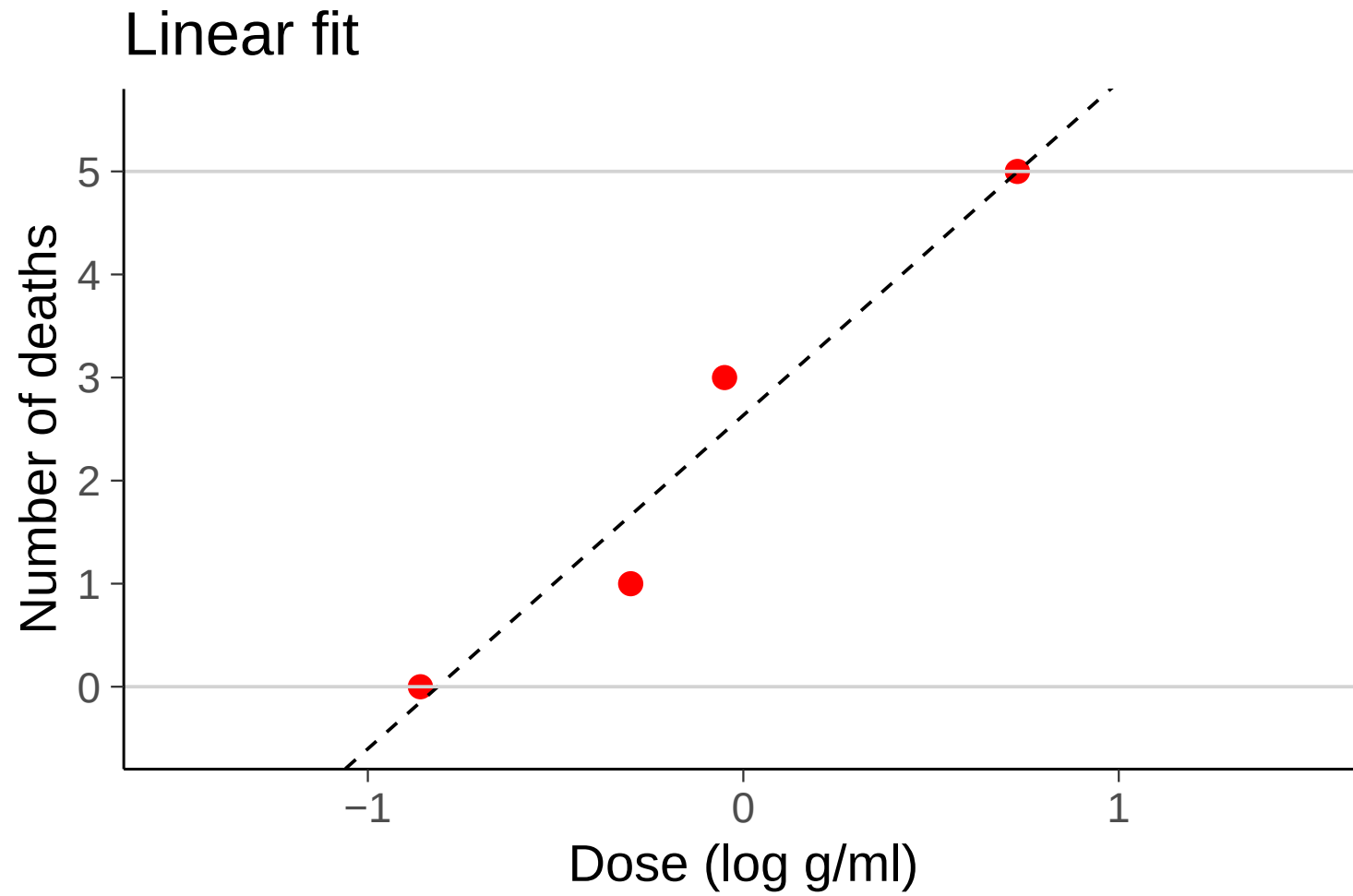
Bioassay



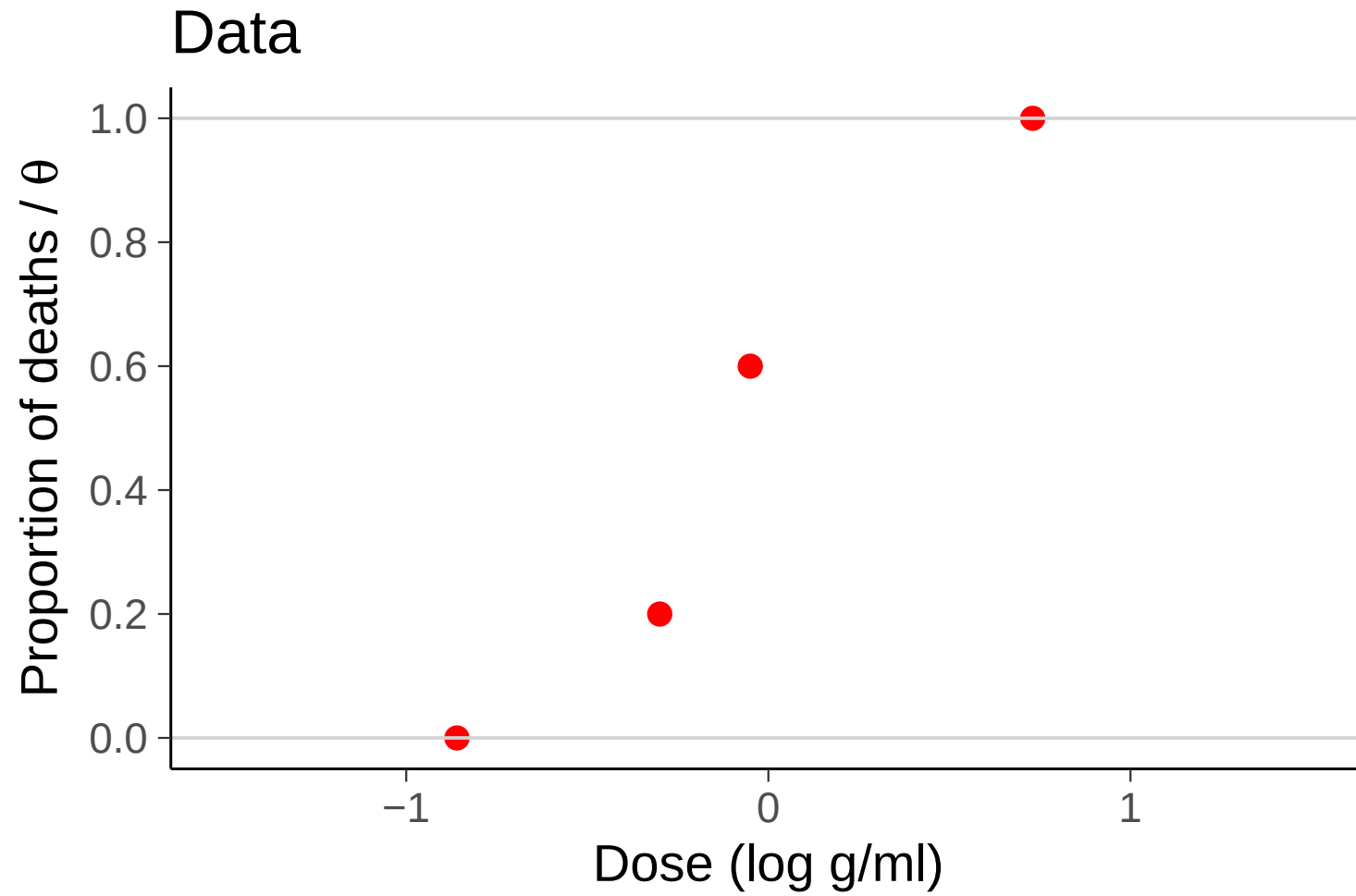
Bioassay



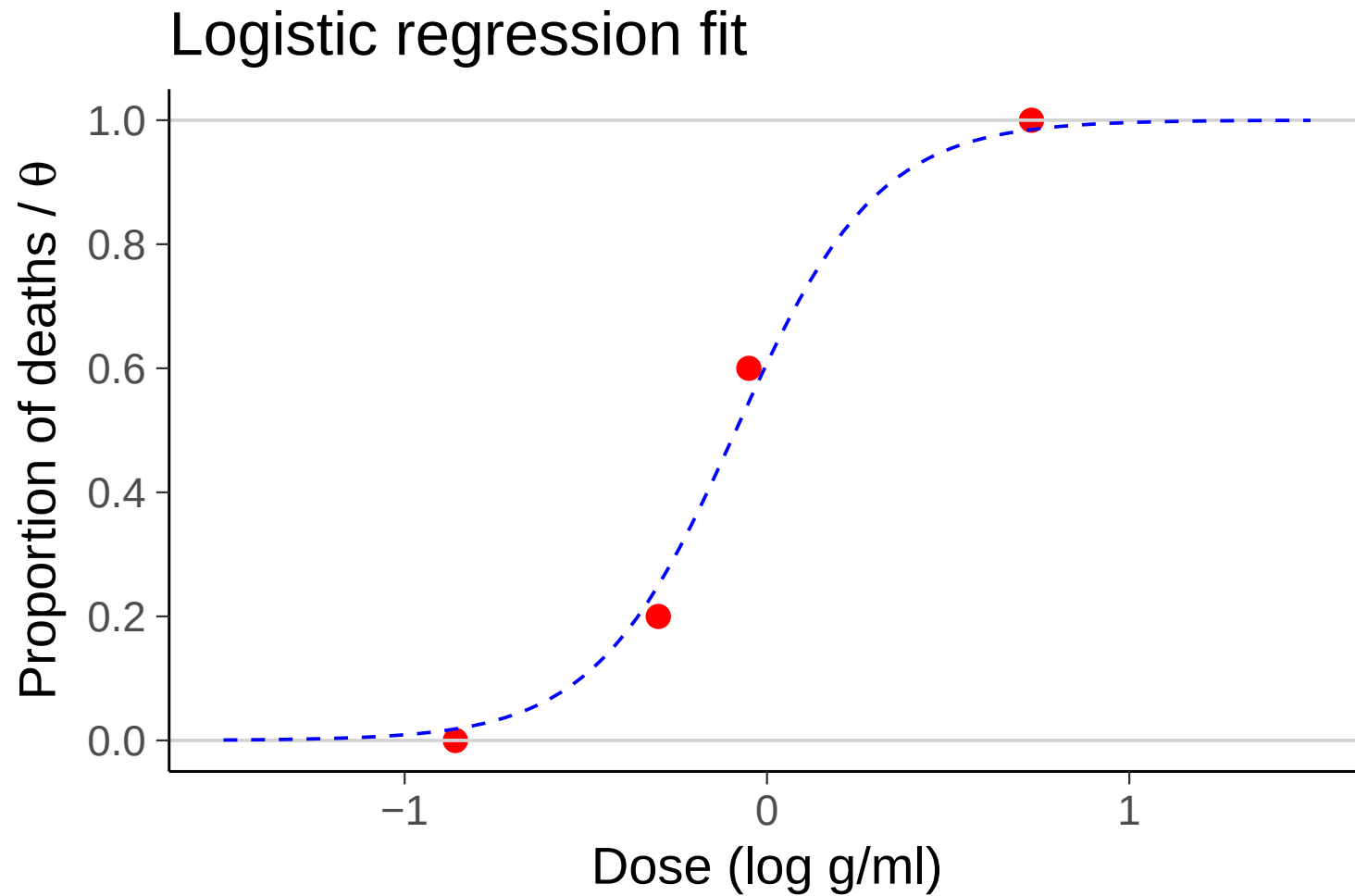
Bioassay



Bioassay



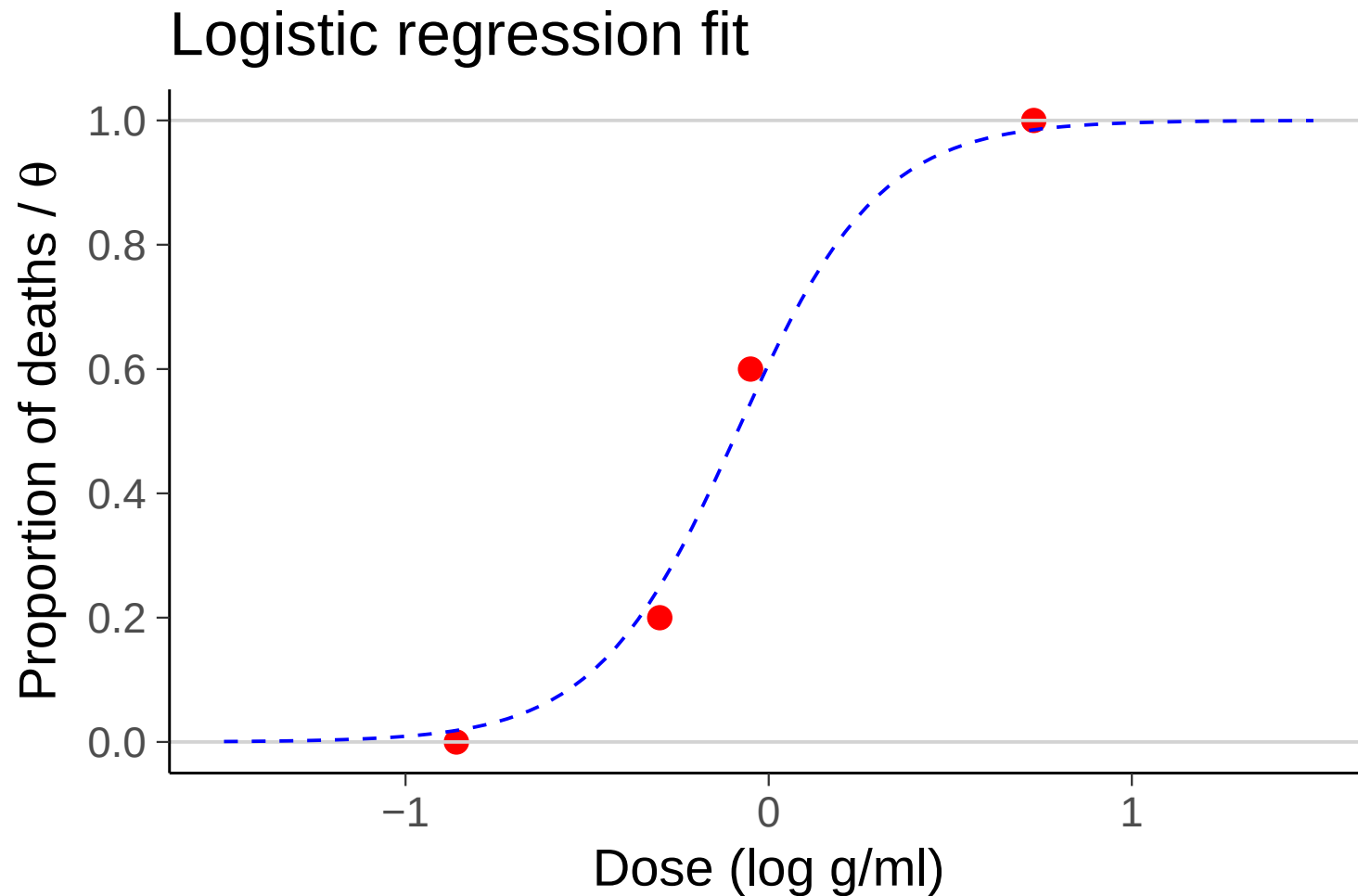
Bioassay



Binomial model

$$y_i \mid \theta_i \sim \text{Bin}(\theta_i, n_i)$$

Bioassay



Binomial model

$$y_i \mid \theta_i \sim \text{Bin}(\theta_i, n_i), \quad \text{logit}(\theta_i) = \log\left(\frac{\theta_i}{1 - \theta_i}\right) = \alpha + \beta x_i$$

Bioassay

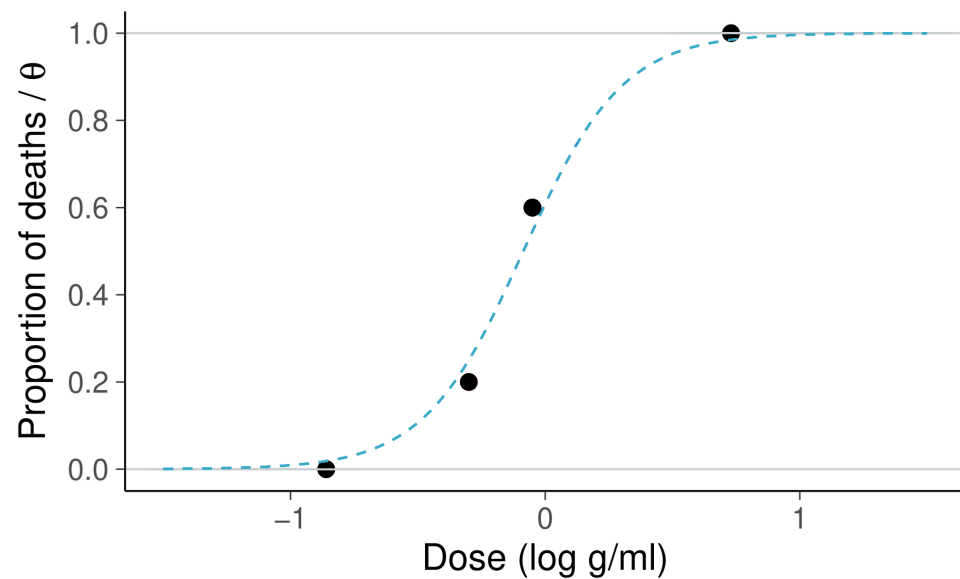
$$y_i \mid \theta_i \sim \text{Bin}(\theta_i, n_i)$$

$$\text{logit}(\theta_i) = \log\left(\frac{\theta_i}{1 - \theta_i}\right)$$

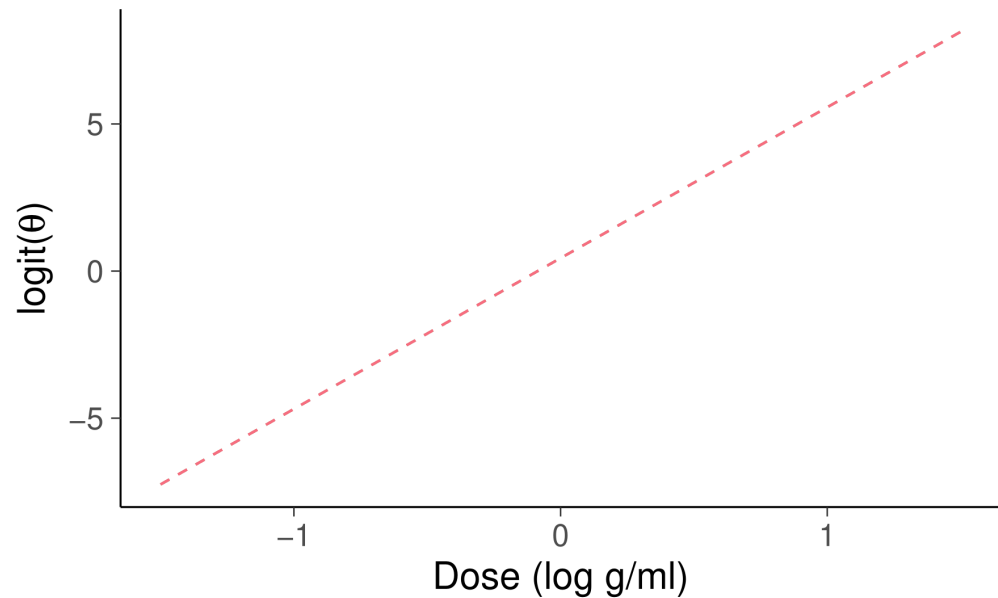
$$= \alpha + \beta x_i$$

$$\theta_i = \frac{1}{1 + \exp(-(\alpha + \beta x_i))}$$

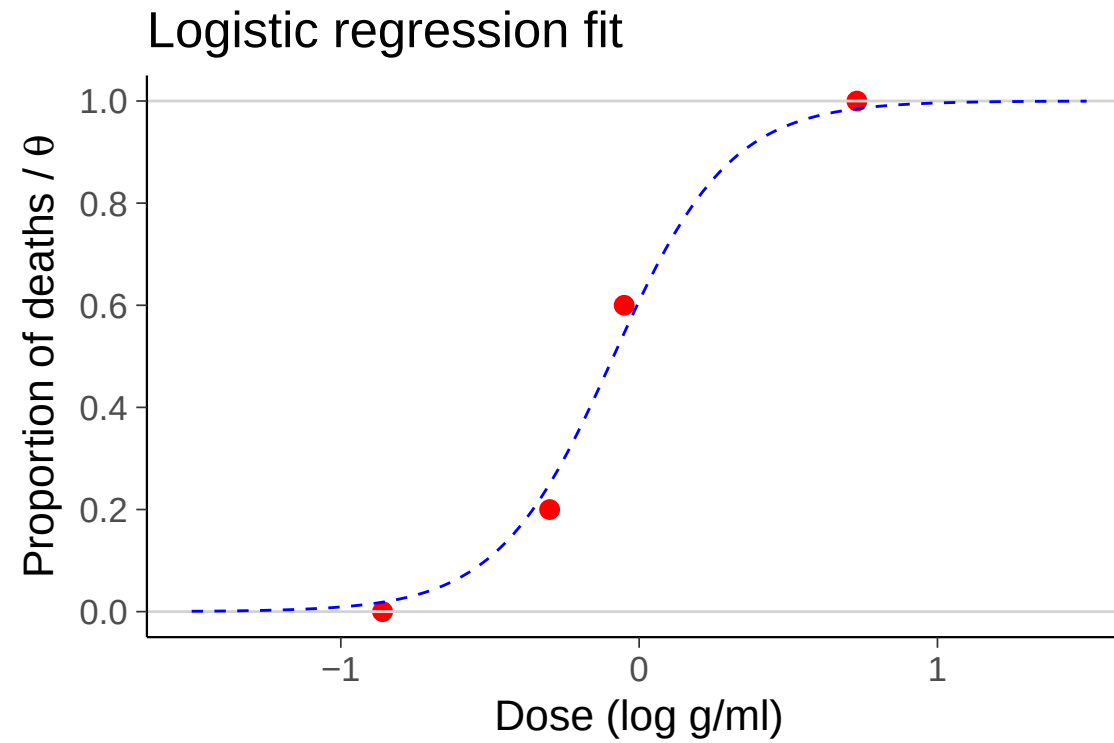
Logistic regression fit



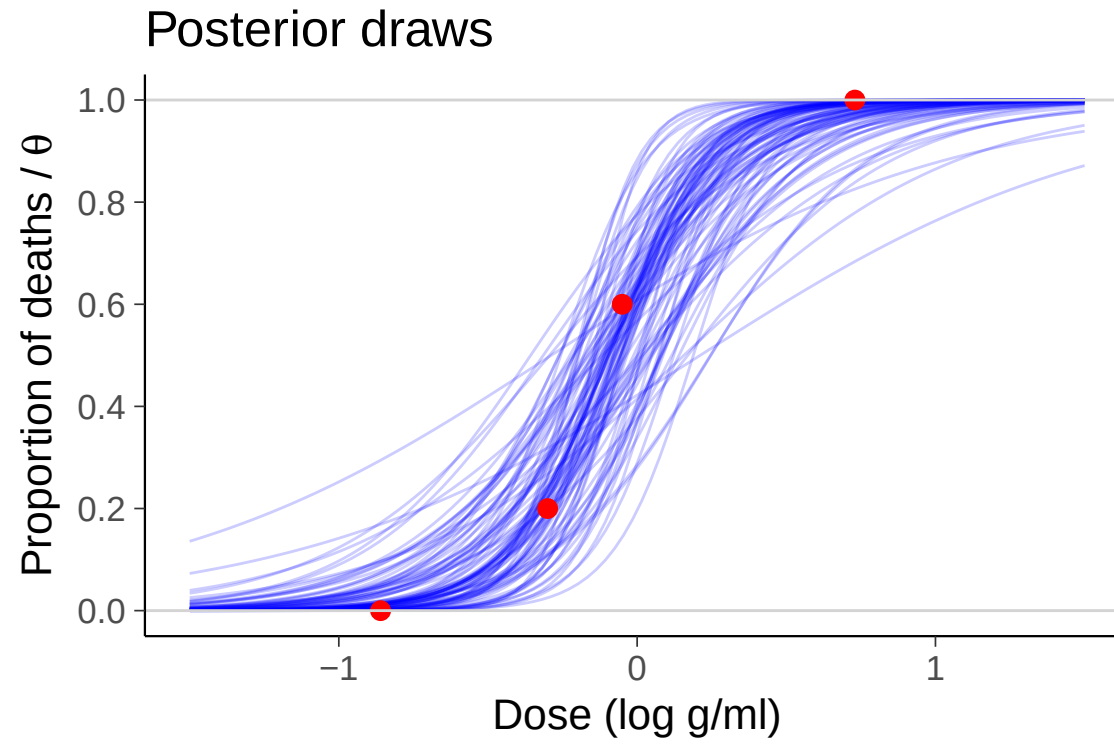
Logistic regression in latent space



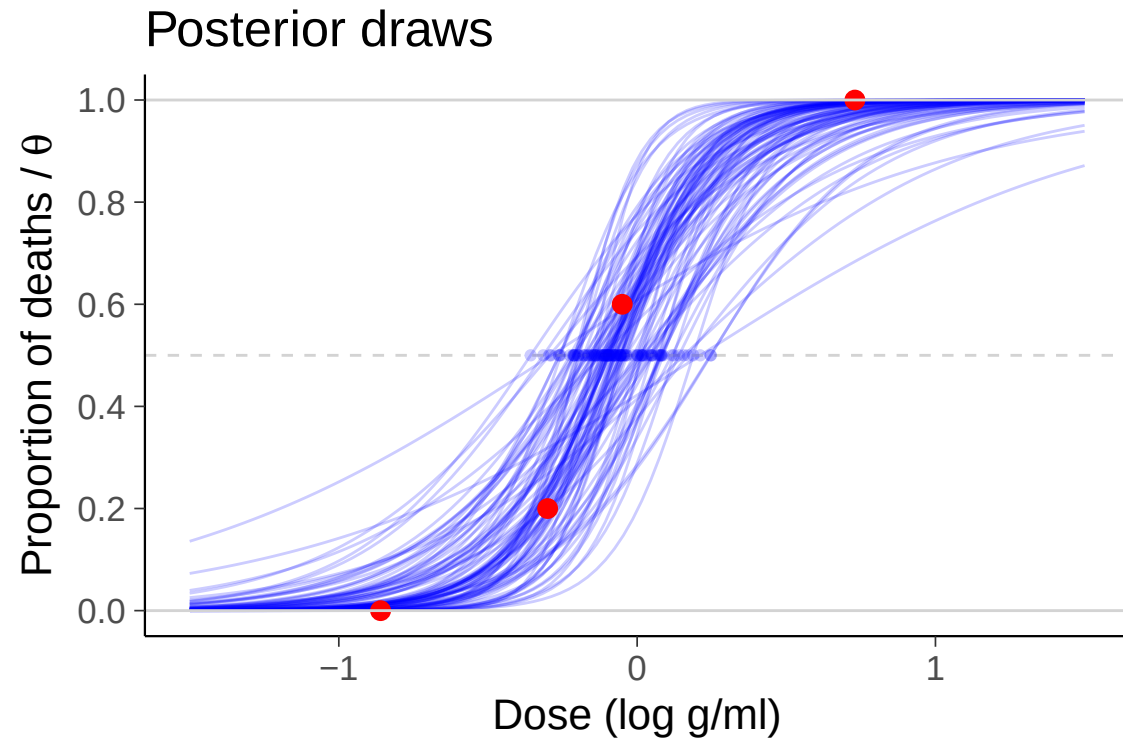
Bioassay



Bioassay

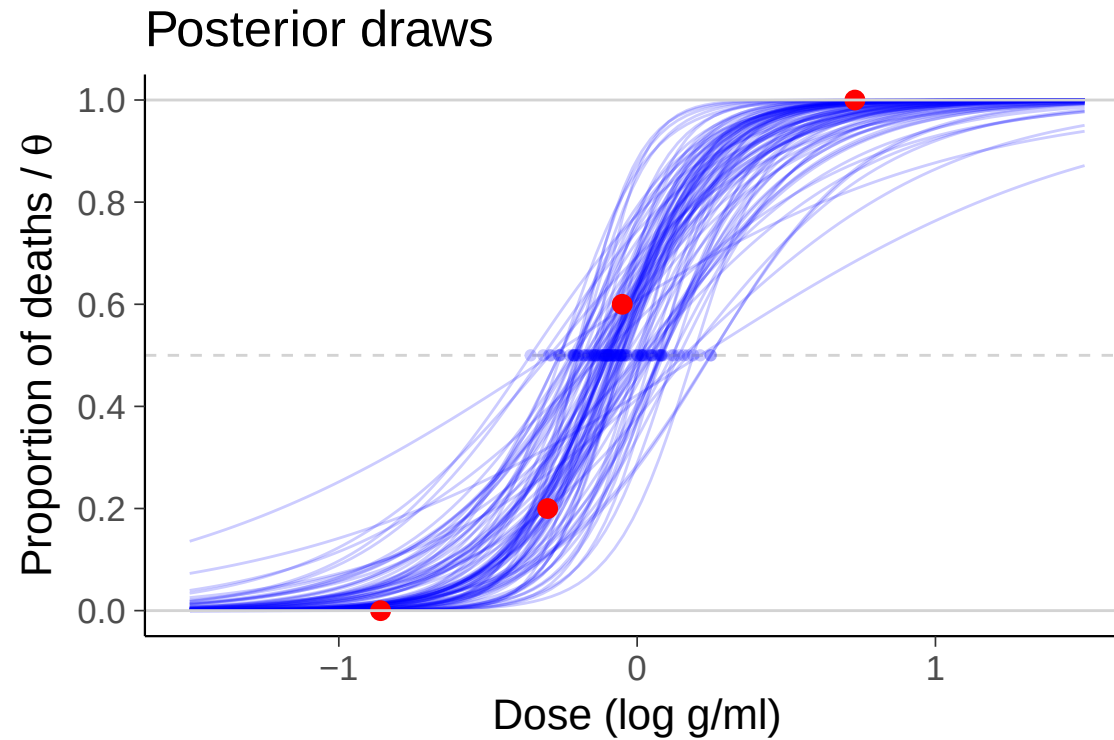


Bioassay



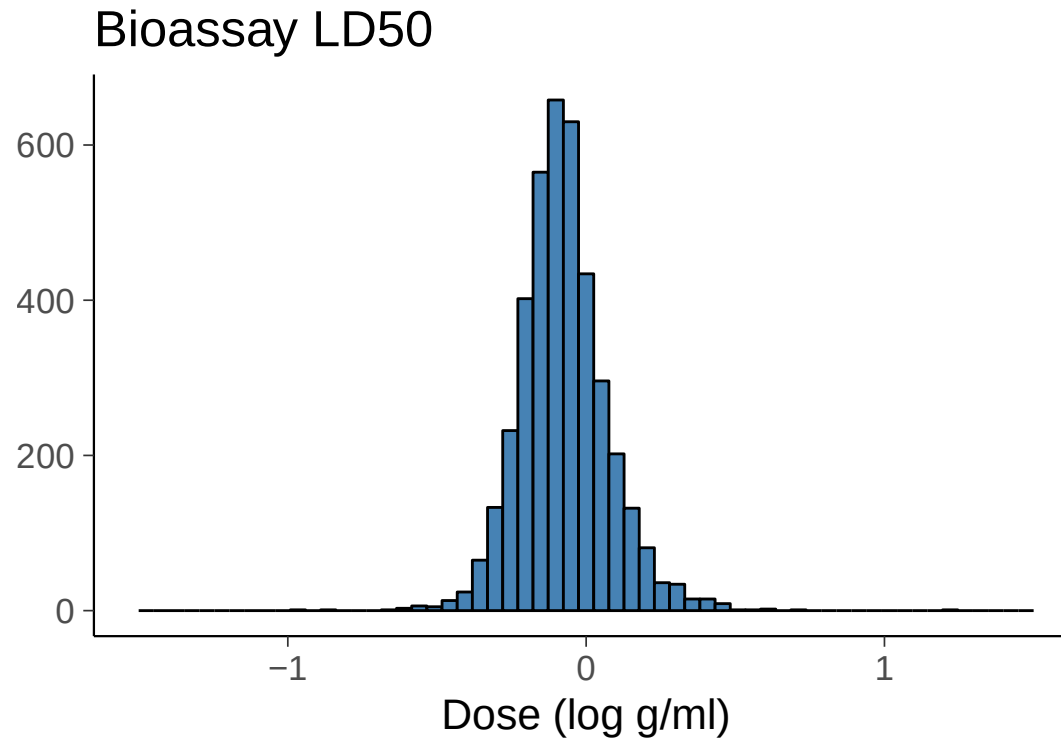
$$\text{LD50: } E[\theta] = \text{logit}^{-1}(\alpha + \beta x) = 0.5 \quad \Rightarrow \quad x_{\text{LD50}} = -\frac{\alpha}{\beta}$$

Bioassay



$$\text{LD50: } E[\theta] = \text{logit}^{-1}(\alpha + \beta x) = 0.5 \Rightarrow x_{\text{LD50}} = -\frac{\alpha}{\beta} \Rightarrow x_{\text{LD50}}^s = -\frac{\alpha^s}{\beta^s}$$

Bioassay



$$\text{LD50: } E[\theta] = \text{logit}^{-1}(\alpha + \beta x) = 0.5 \quad \Rightarrow \quad x_{\text{LD50}} = -\frac{\alpha}{\beta} \quad \Rightarrow \quad x_{\text{LD50}}^s = -\frac{\alpha^s}{\beta^s}$$

Bioassay posterior

Binomial model

$$y_i \mid \theta_i \sim \text{Bin}(\theta_i, n_i)$$

Link function

$$\text{logit}(\theta_i) = \alpha + \beta x_i$$

Likelihood

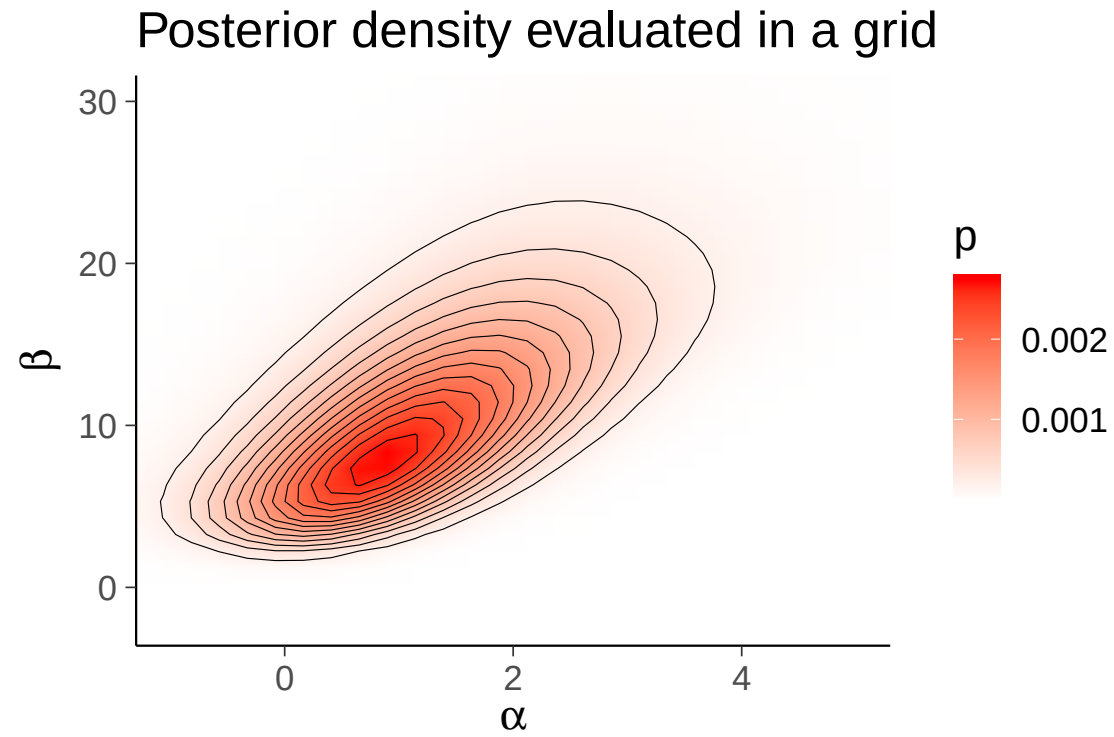
$$p(y_i \mid \alpha, \beta, n_i, x_i) \propto \theta_i^{y_i} [1 - \theta_i]^{n_i - y_i}$$

$$p(y_i \mid \alpha, \beta, n_i, x_i) \propto [\text{logit}^{-1}(\alpha + \beta x_i)]^{y_i} [1 - \text{logit}^{-1}(\alpha + \beta x_i)]^{n_i - y_i}$$

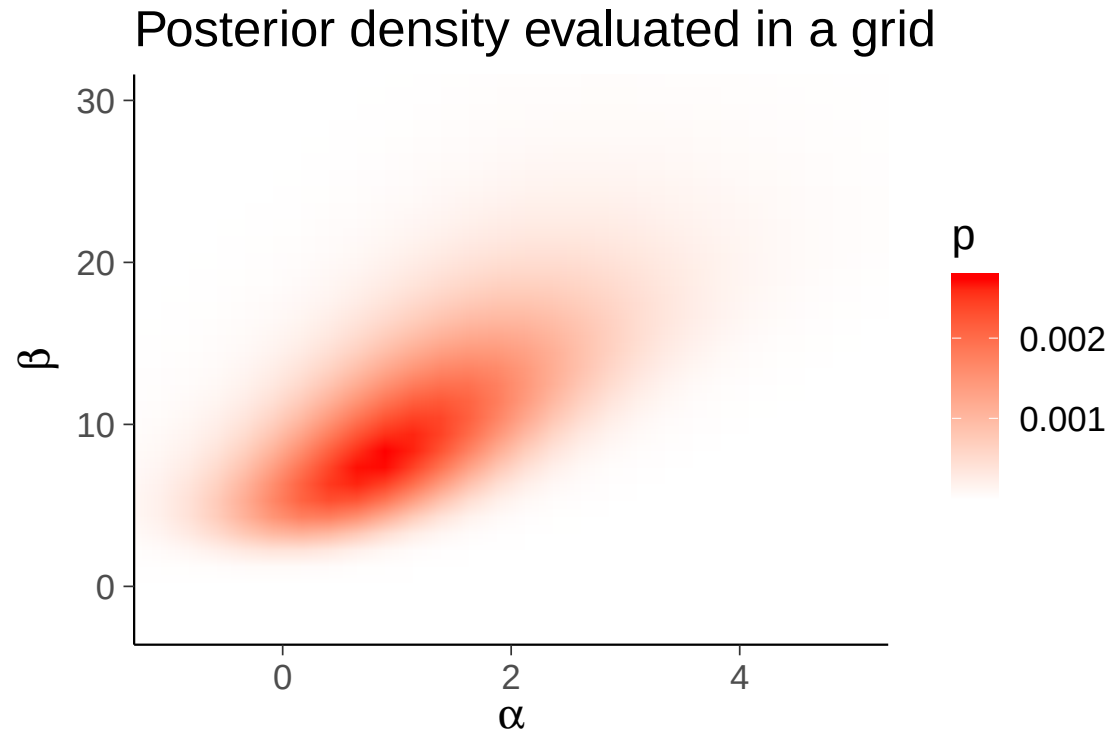
Posterior (with uniform prior on α, β)

$$p(\alpha, \beta \mid y, n, x) \propto p(\alpha, \beta) \prod_{i=1}^n p(y_i \mid \alpha, \beta, n_i, x_i)$$

Bioassay - grid

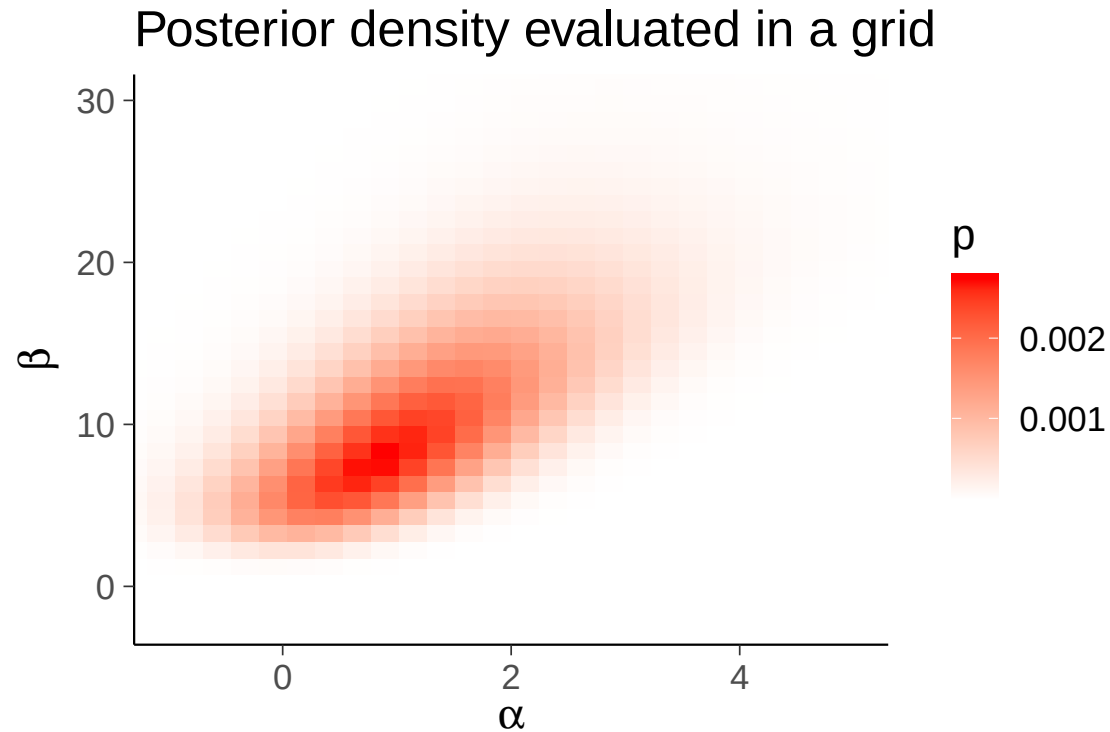


Bioassay - grid



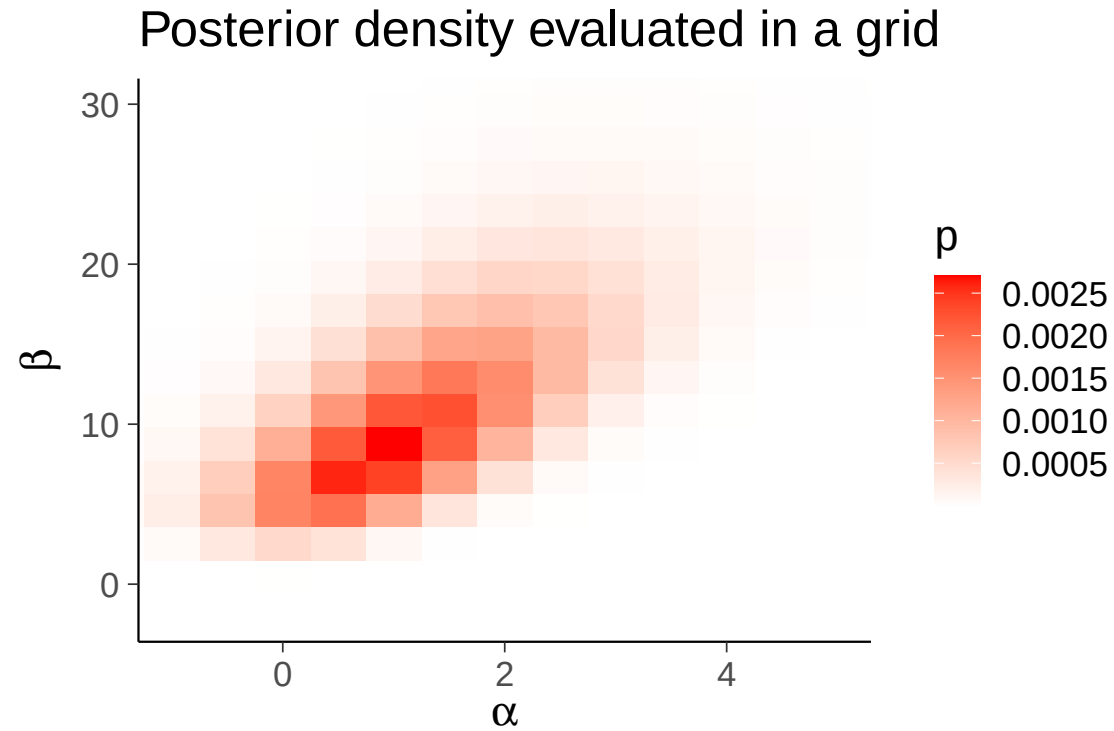
Density evaluated in grid, but plotted using interpolation

Bioassay - grid



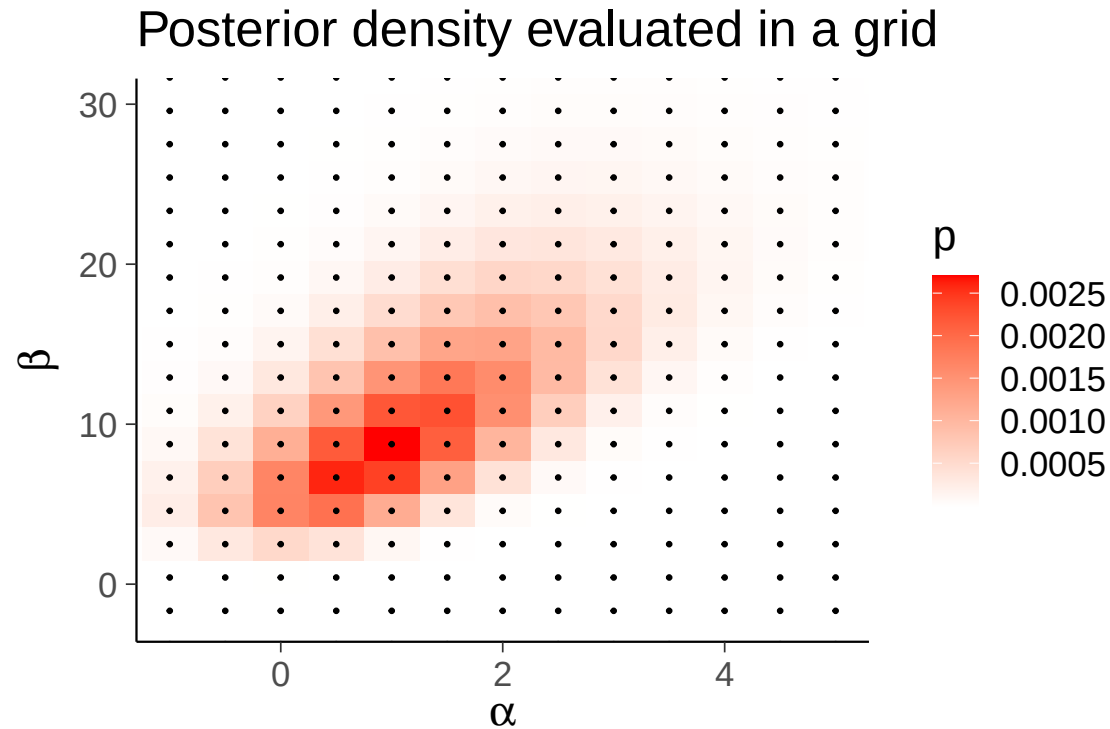
Density evaluated in grid, and plotted without interpolation

Bioassay - grid



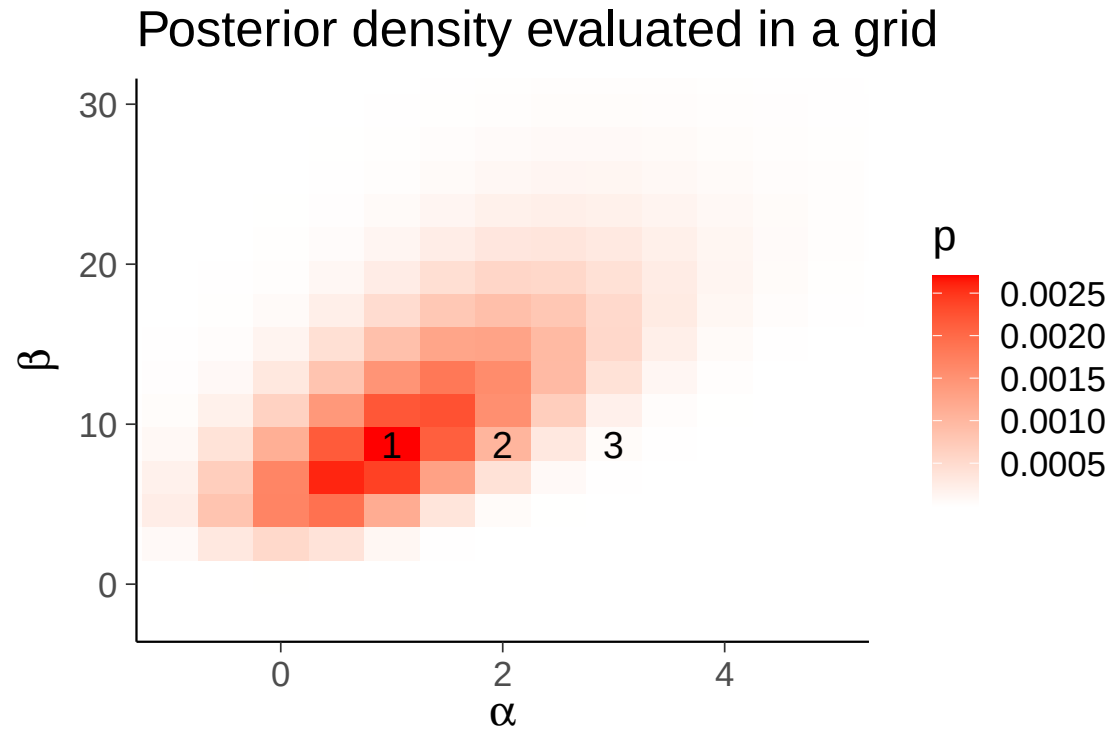
Density evaluated in a coarser grid

Bioassay - grid



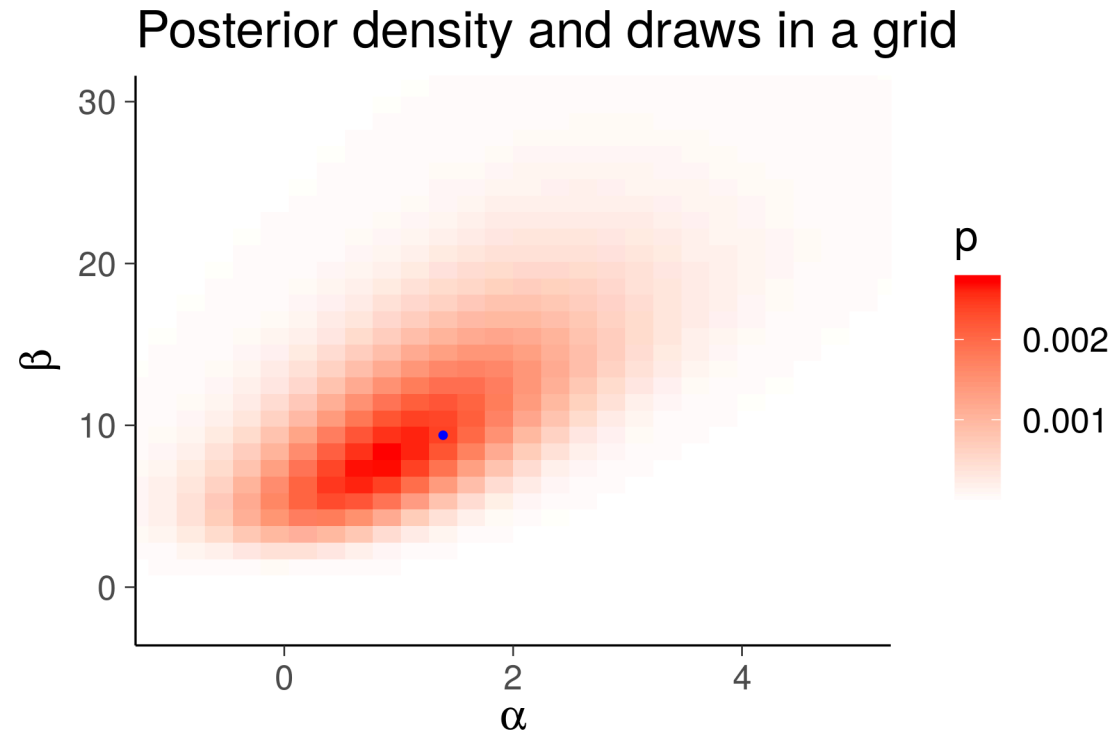
- Approximate the density as piecewise constant function
- Evaluate density in a grid over some finite region
- Density times cell area gives probability mass in each cell

Bioassay - grid

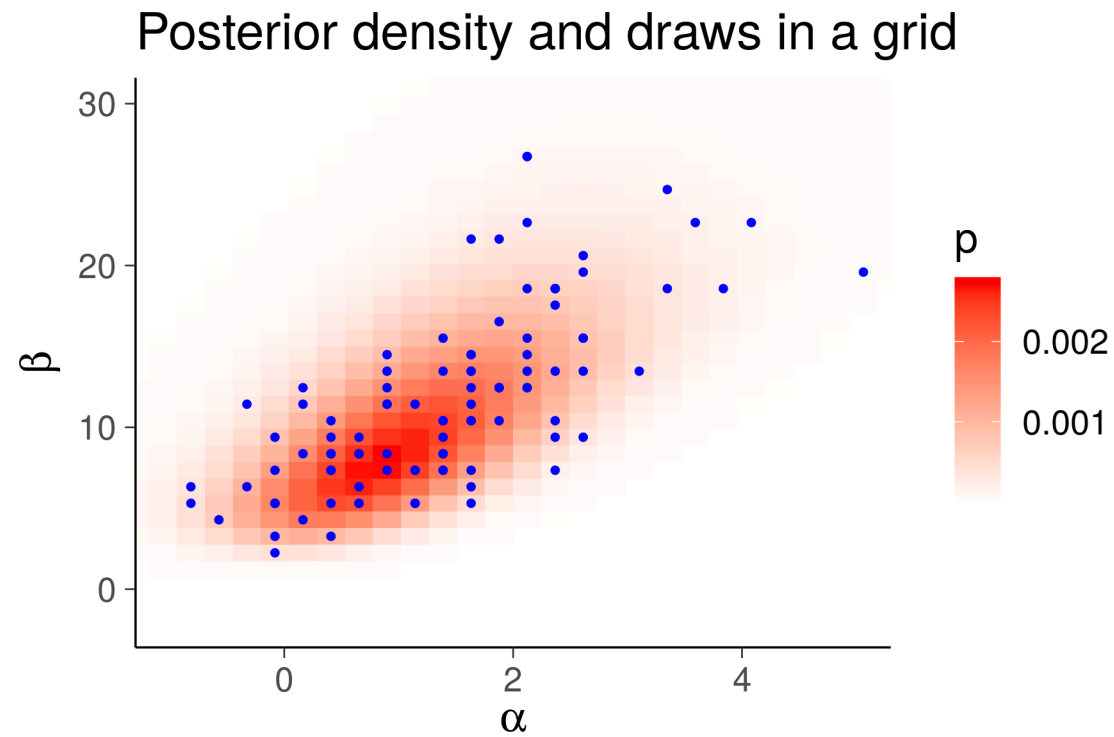


- Densities at 1, 2, and 3: 0.0027 0.0010 0.0001
- Probabilities of cells 1, 2, and 3: 0.0431 0.0166 0.0010
- Probabilities of cells sum to 1

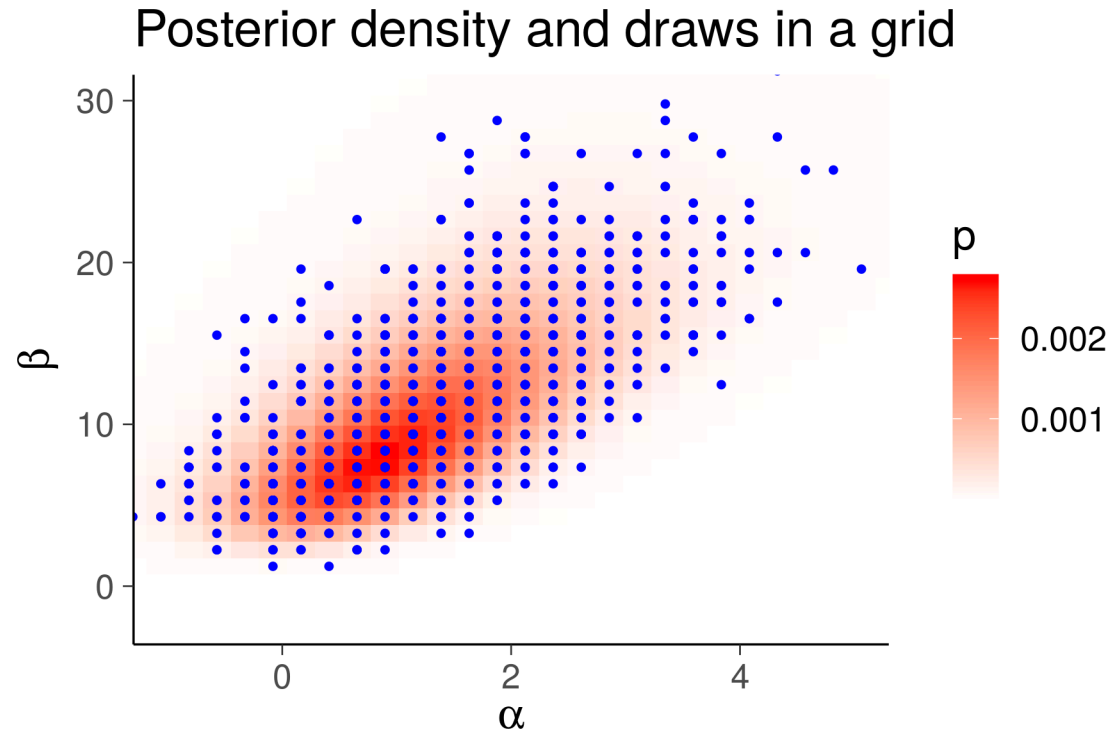
Bioassay - grid



Bioassay - grid

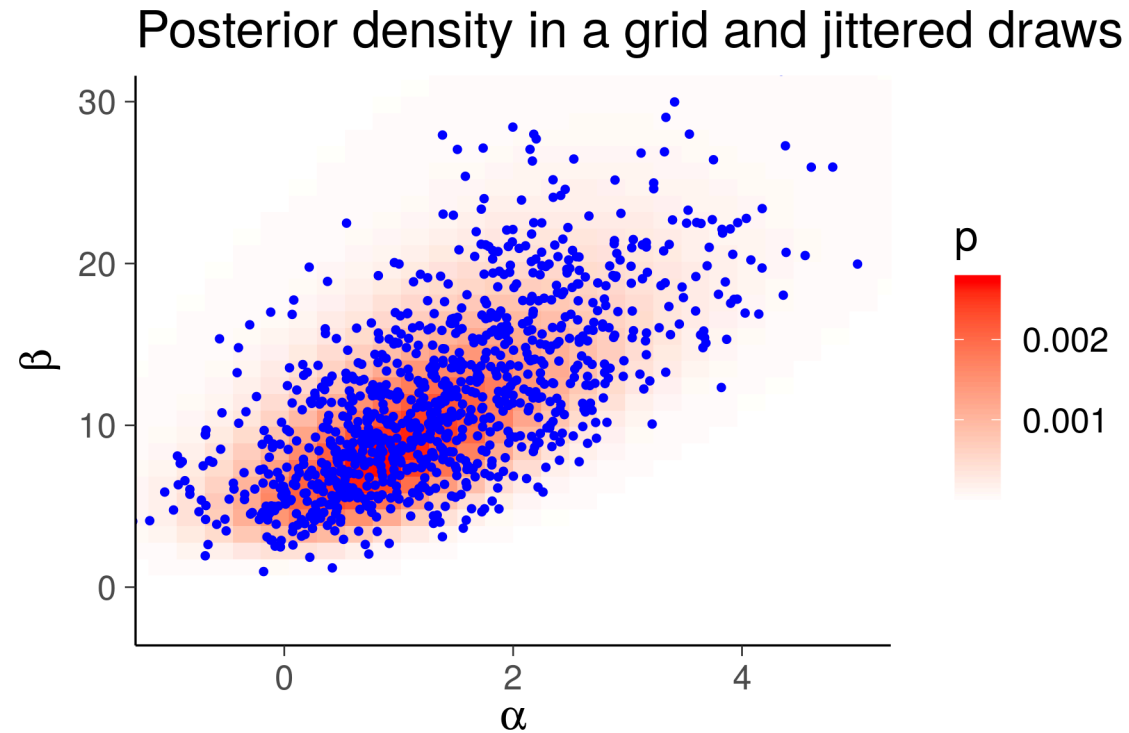


Bioassay - grid



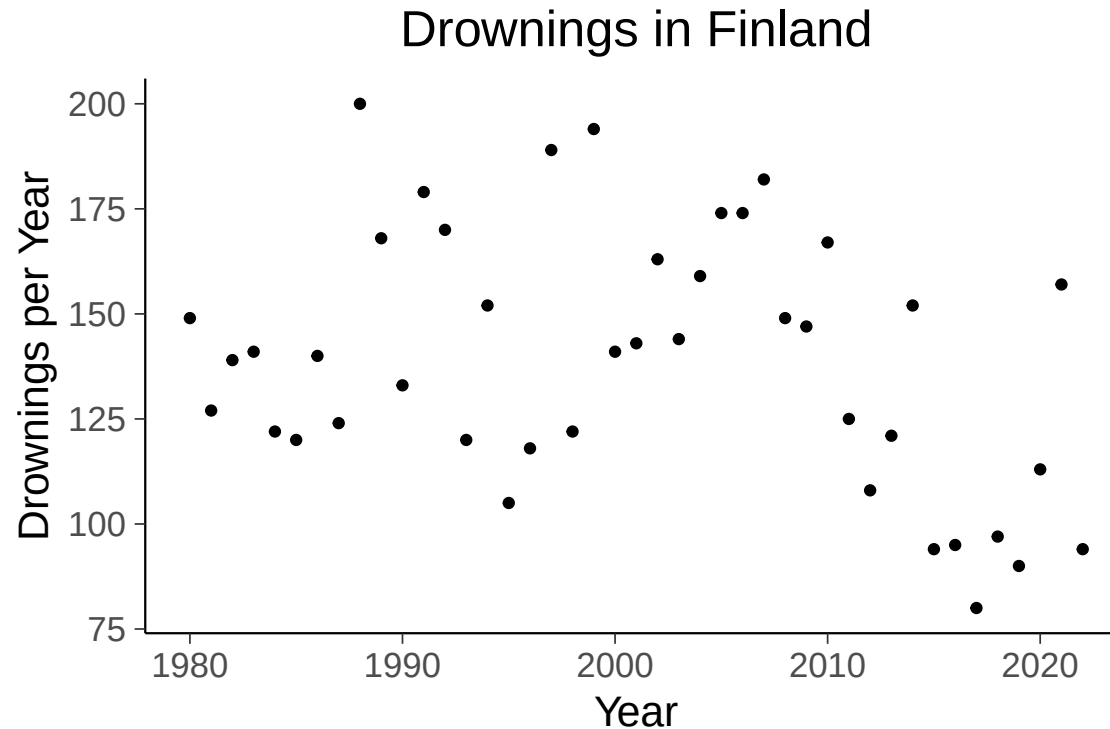
- Sample according to grid cell probabilities
- Several draws can be from the same grid cell

Bioassay - grid



- Jitter can be added to improve visualization

Example: GLM for counts



- On average ~140 drownings per year
- Finnish government has invested in measures for reducing deaths
- Recent narrative based on effectiveness of education

Bayesian methods help

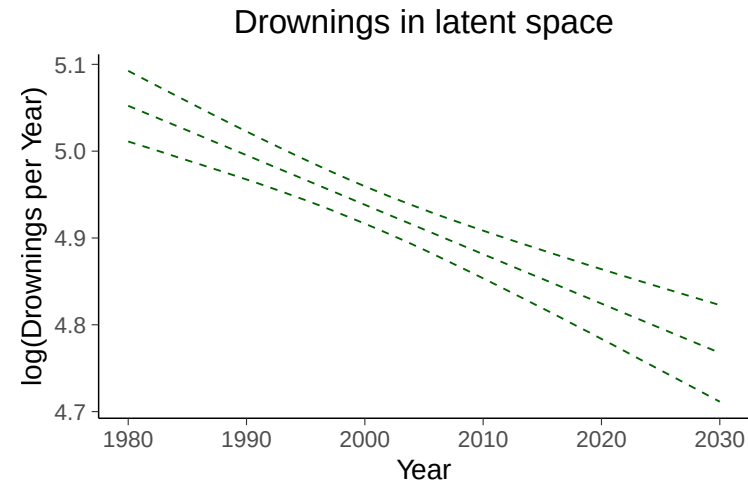
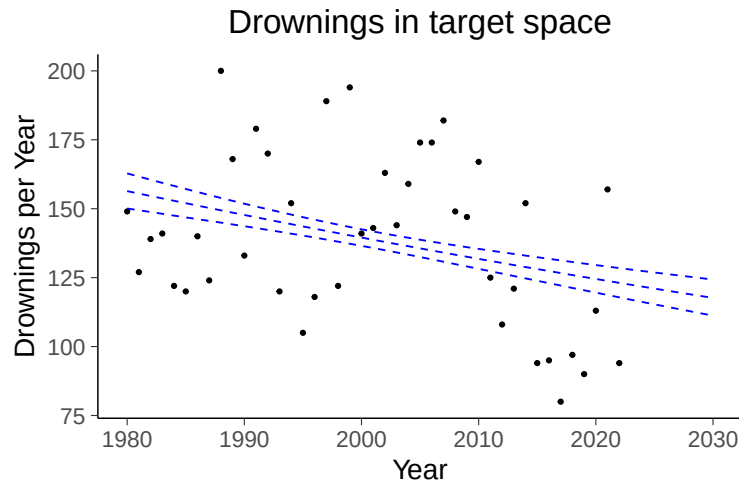
- Describe trends over time
- Evaluate uncertainty

Example: GLM for counts

- Poisson regression model for counts:

$$y_i \mid \mu_i \sim \text{Poisson}(\mu_i)$$

$$\mu_i = e^{\alpha + \beta x_i}$$

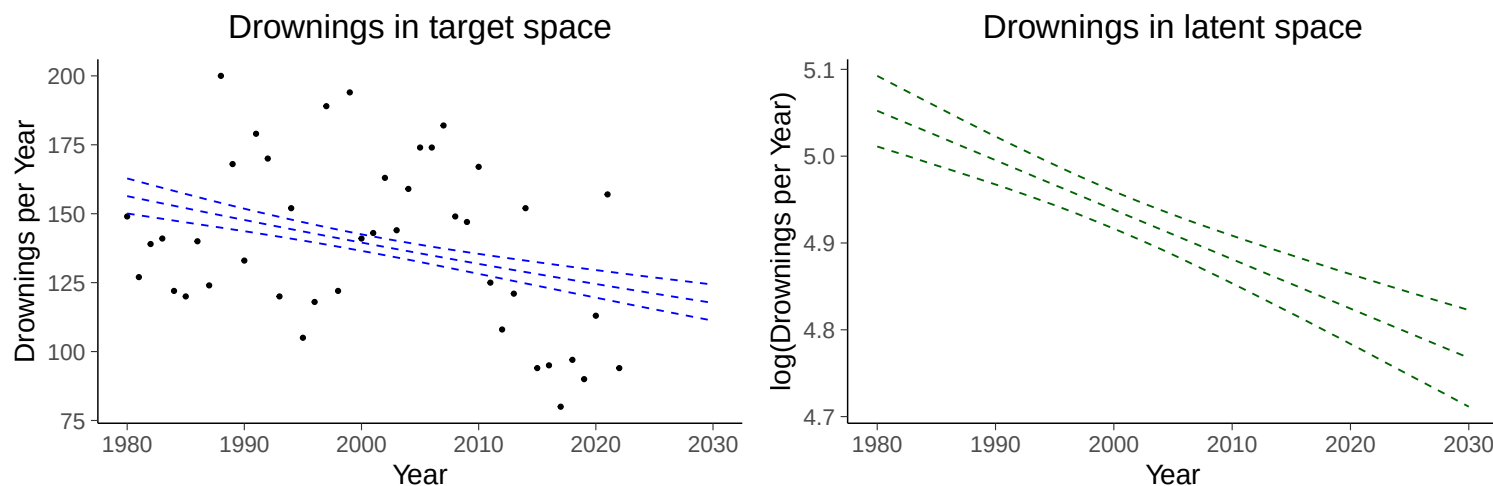


Example: GLM for counts

- Poisson regression model for counts:

$$y_i \mid \mu_i \sim \text{Poisson}(\mu_i)$$

$$\mu_i = e^{\alpha + \beta x_i}$$



- Alternatively, Negative Binomial regression can be used for overdispersed count data.

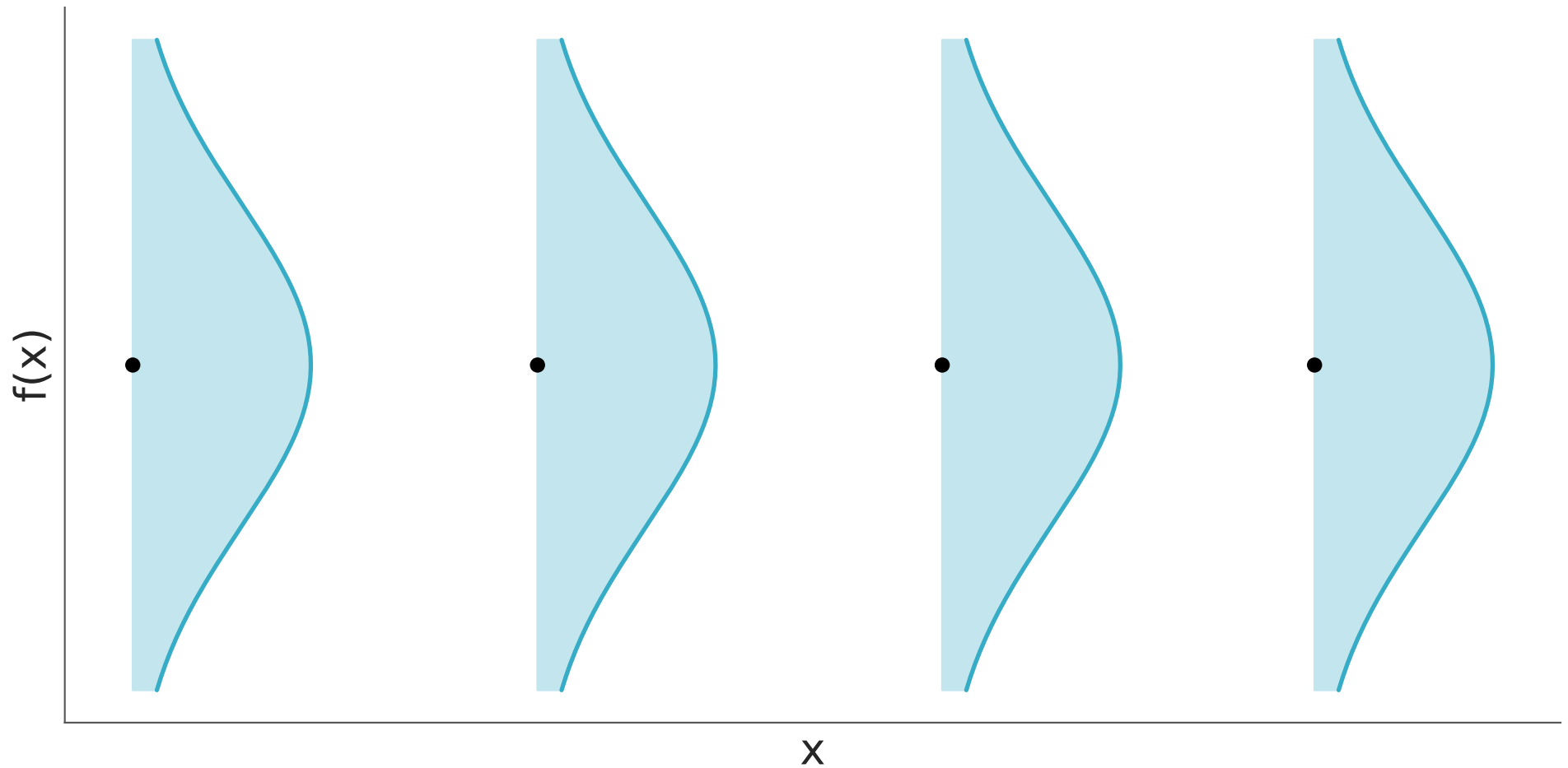
$$y_i \mid \mu_i, \varphi \sim \text{Neg-bin}(y_i \mid \mu_i, \varphi)$$

Gaussian processes

- Imagine we want to represent a function in probabilistic terms

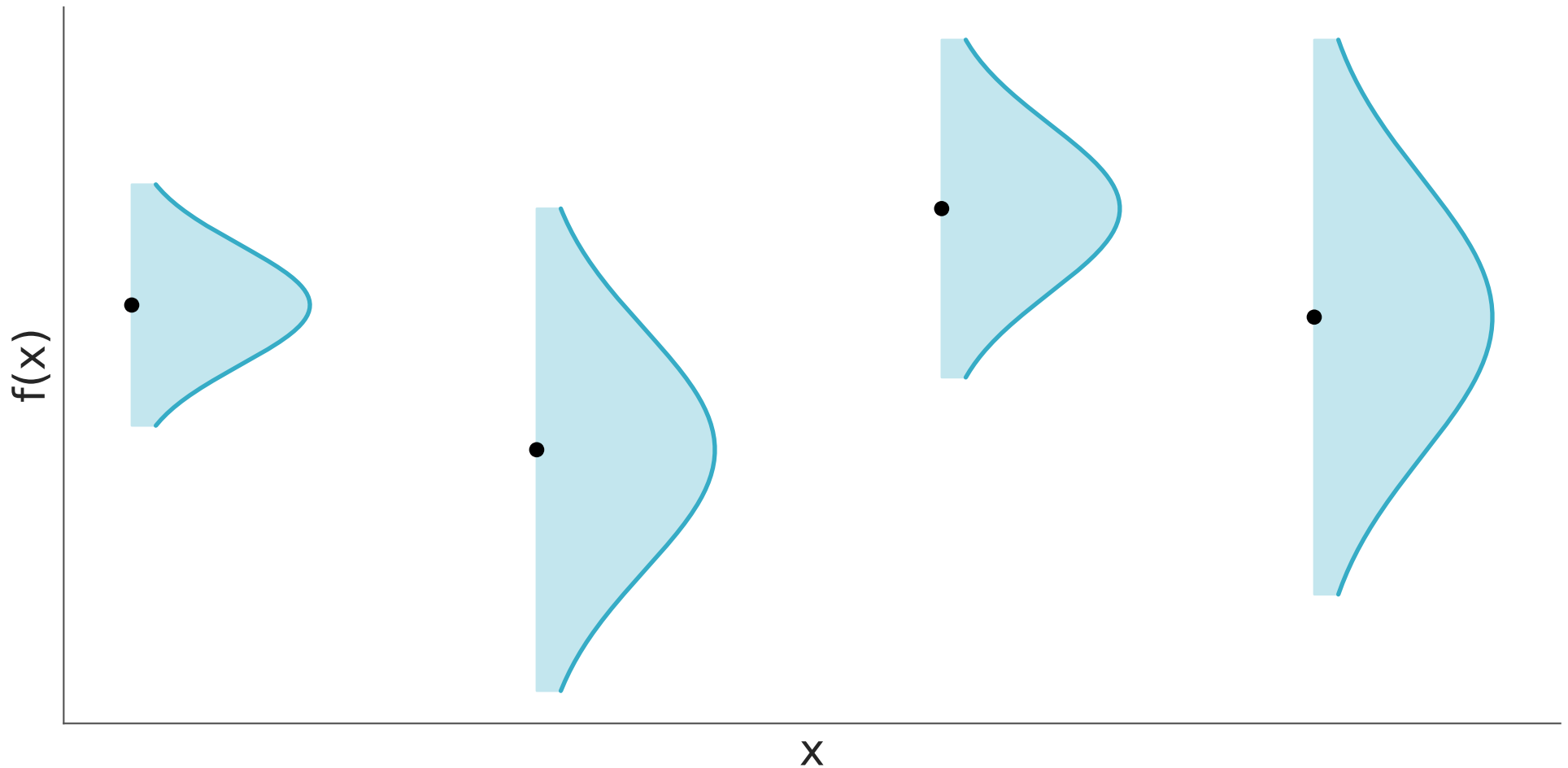
Gaussian processes

- Imagine we want to represent a function in probabilistic terms
- We could associate a Gaussian with each function value.



Gaussian processes

- We can vary the mean and variance of the Gaussian associated with each function value.

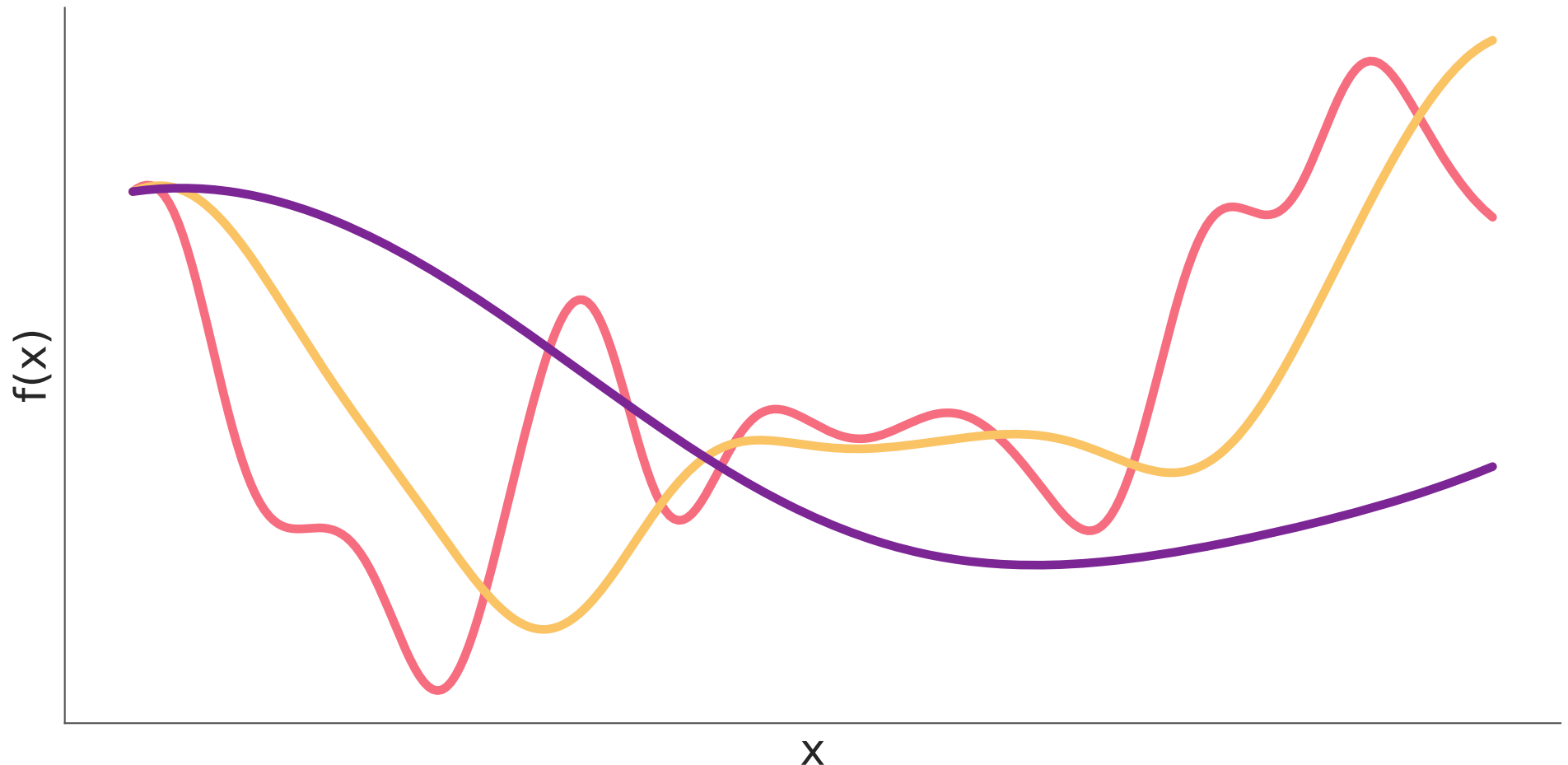


Gaussian processes

- Instead of independent normals, we use a Multivariate normal distribution to model the joint behavior of all function values.

Gaussian processes

- Instead of independent normals, we use a Multivariate normal distribution to model the joint behavior of all function values.
- This way, we introduce correlations between function values at different points.



Gaussian processes

- Formally, it's an infinite-dimensional multivariate normal - any finite set of function values, $f(x_1), \dots, f(x_n)$, is jointly normal.

Gaussian processes

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- In practice we only ever evaluate a function at **finitely** many points, so we only ever need this finite-Multivariate normal

Gaussian processes

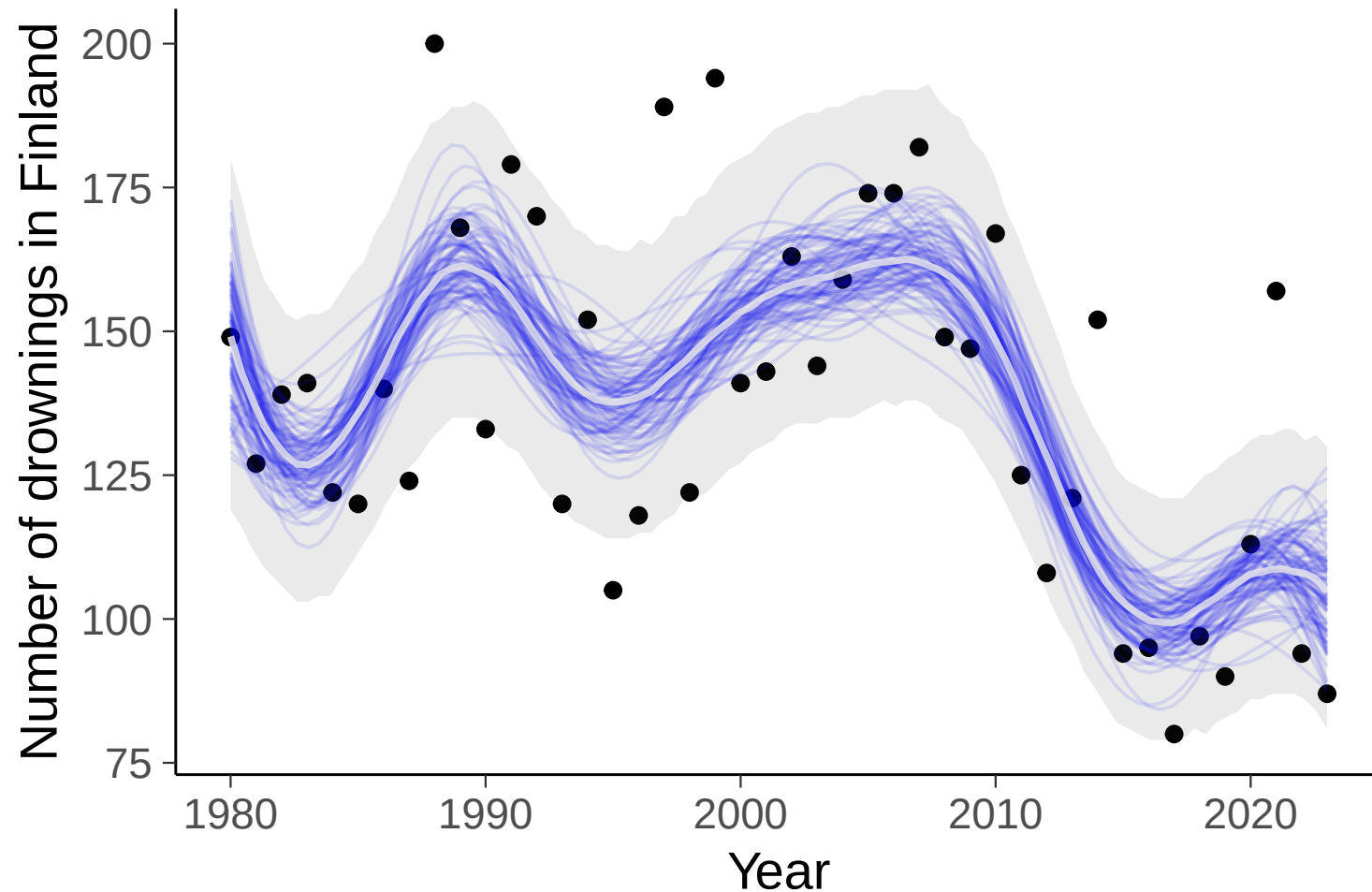
- Formally, it's an infinite-dimensional multivariate normal - any finite set of function values, $f(x_1), \dots, f(x_n)$, is jointly normal.
- In practice we only ever evaluate a function at **finitely** many points, so we only ever need this finite-Multivariate normal
- Some nice theoretical properties and many models can be seen as as special cases of Gaussian processes.

Gaussian processes

- Formally, it's an infinite-dimensional multivariate normal - any finite set of function values, $f(x_1), \dots, f(x_n)$, is jointly normal.
- In practice we only ever evaluate a function at **finitely** many points, so we only ever need this finite-Multivariate normal
- Some nice theoretical properties and many models can be seen as as special cases of Gaussian processes.
- Used for regression/classification when you want a flexible, non-parametric function with built-in uncertainty.

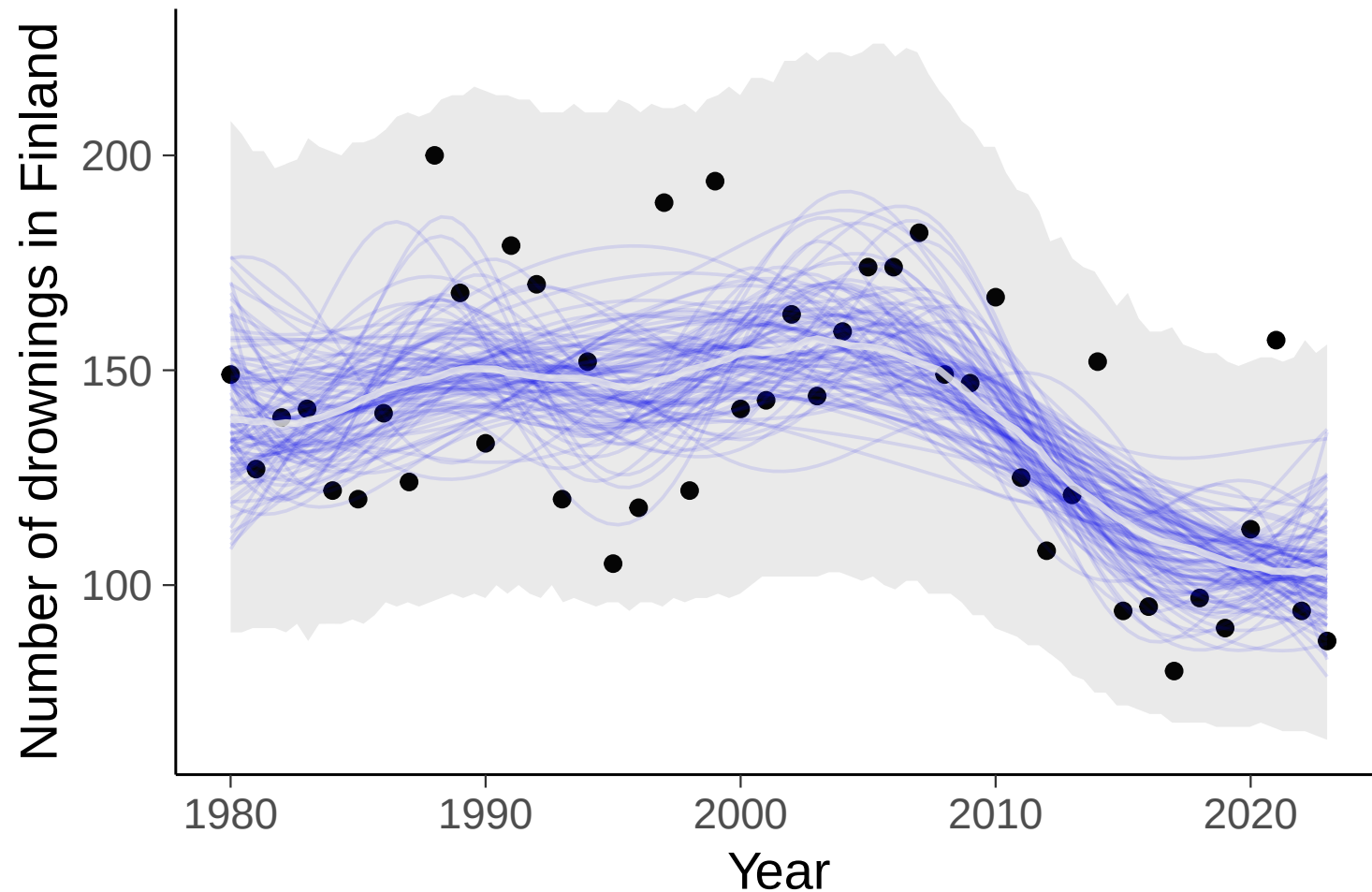
Example GLM: Gaussian Process Models

$$y_i \mid \mu_i \sim \text{Poisson}(\mu_i) \quad \mu_i \sim e^{f_i}, \quad f \sim \text{GP}(0, k(\text{Year}, \theta))$$



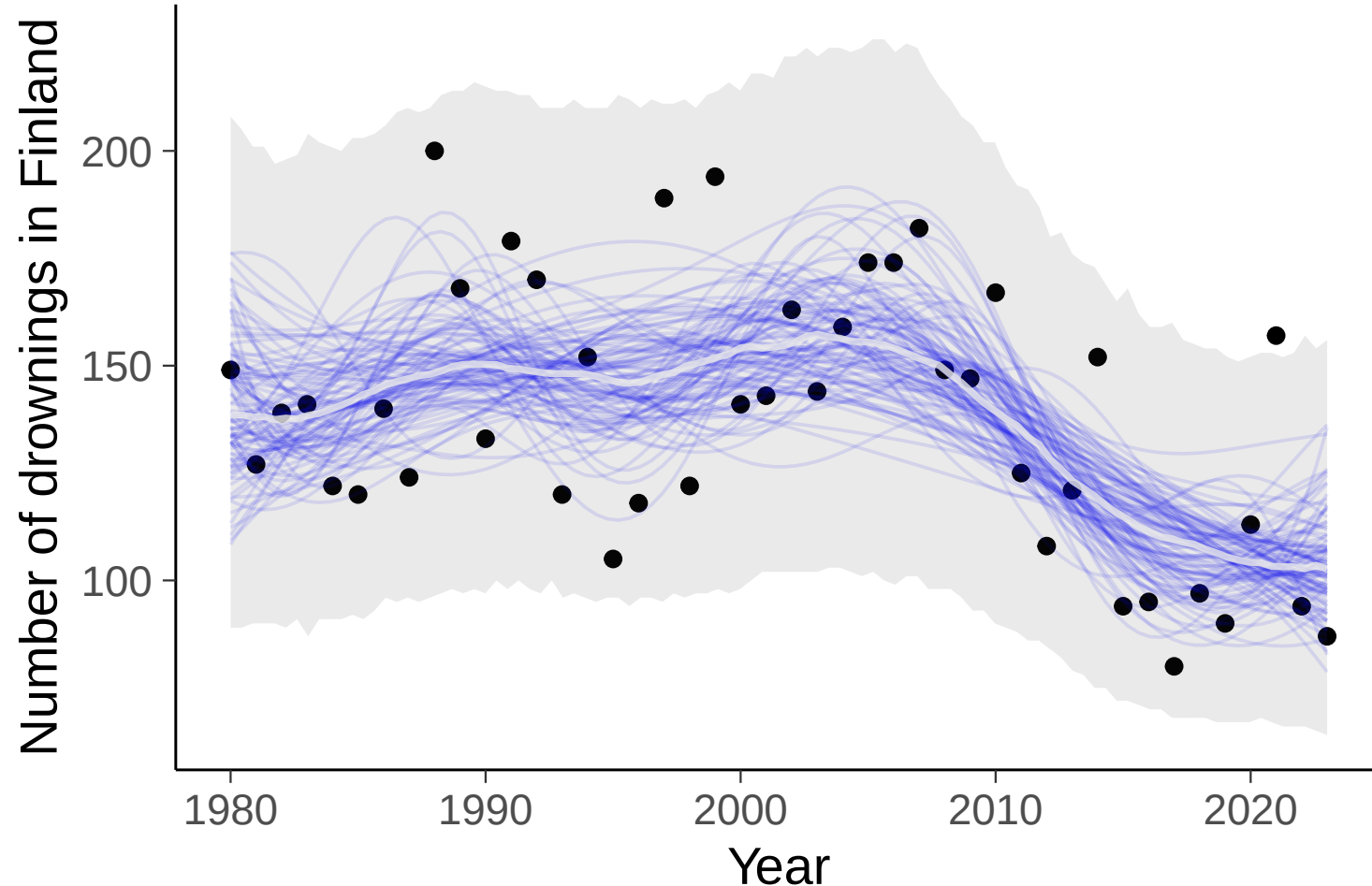
Example GLM: Gaussian Process Models

$$y_i \mid \mu_i \sim \text{Neg-bin}(\mu_i, \varphi) \quad \mu_i \sim e^{f_i}, \quad f \sim \text{GP}(0, k(\text{Year}, \theta))$$



Example GLM: Gaussian Process Models

- Clear overdispersion
 - later we use posterior predictive checking and cross-validation to confirm this
- Trend interpretations shouldn't be based on one observation



Thinking counts

- For simplicity of exposition, we often start learning with normal observation models
- But we observe count data on a daily basis
- Very relevant in industry (number of sold products, ad views, customer count, etc.)
- Can you think of such examples from the classroom?
 - Think of how many students attend BDA lectures over the course
 - Number of students who report getting sick over time until Christmas
 - Number of dropouts
 - Would you expect overdispersion?

Appendix A1: derivation of the sum identity

We now show the identity used above, $\sum_i (y_i - \mu)^2 = \sum_i (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2$, by expanding the square:

$$\begin{aligned} \sum_{i=1}^n (y_i - \mu)^2 \\ = \sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2) \end{aligned}$$

We add and subtract $\bar{y}^2 - 2y_i\bar{y}$ inside the sum — this doesn't change the value, but lets us regroup terms around $\bar{y}$ instead of μ :

$$= \sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2 - \bar{y}^2 + \bar{y}^2 - 2y_i\bar{y} + 2y_i\bar{y})$$

Grouping the terms that involve only y_i and $\bar{y}$ into one sum, and the remaining terms into another:

$$= \sum_{i=1}^n (y_i^2 - 2y_i\bar{y} + \bar{y}^2) + \sum_{i=1}^n (\mu^2 - 2y_i\mu - \bar{y}^2 + 2y_i\bar{y})$$

Appendix A2: Marginal posterior $p(\mu \mid y)$ derivation

We now derive $p(\mu \mid y)$ analytically, by integrating the joint posterior over σ^2 :

$$\begin{aligned} p(\mu \mid y) &= \int_0^\infty p(\mu, \sigma^2 \mid y) d\sigma^2 \\ &\propto \int_0^\infty \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2]\right) d\sigma^2 \end{aligned}$$

To evaluate this, substitute a new variable z that absorbs the bracket term and turns the integral into a recognizable form:

Transformation

$$A = (n-1)s^2 + n(\mu - \bar{y})^2 \quad \text{and} \quad z = \frac{A}{2\sigma^2}$$

$$p(\mu \mid y) \propto A^{-\frac{n}{2}} \int_0^\infty z^{\frac{n-2}{2}} \exp(-z) dz$$

Recognize gamma integral $\Gamma(u) \equiv \int_0^\infty x^{u-1} \exp(-x) dx$



Appendix A2: Marginal posterior $p(\mu \mid y)$

(cont)

The integral is now just a constant (a gamma function value not depending on μ), leaving only A 's dependence on μ :

$$\propto [(n-1)s^2 + n(\mu - \bar{y})^2]^{-\frac{n}{2}}$$

Dividing through by $(n-1)s^2$ to isolate the standard Student- t kernel form:

$$\propto \left[1 + \frac{n(\mu - \bar{y})^2}{(n-1)s^2} \right]^{-\frac{n}{2}}$$

$$p(\mu \mid y) = t_{n-1}\left(\mu \mid \bar{y}, \frac{s^2}{n}\right) \text{ Student's } t$$